

# MICROSAR OSEK Direct Network Management

Technical Reference

Complex Device Driver  
Version 7.00.00

|         |                                         |
|---------|-----------------------------------------|
| Authors | Markus Drescher, Leticia Garcia Herrera |
| Status  | Released                                |

## Document Information

### History

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| Leticia Garcia Herrera | 2017-02-10 | 6.00.00 | FEATC-870 Adapted chapters 2.8.3 and 4.3; added chapter 2.10.12                                                                                                                                                                                                                                                                       |
| Leticia Garcia Herrera | 2017-08-30 | 7.00.00 | FEAT-2482 Adapted chapters 2.5.1, 2.8.1 and 4.1 adapted<br>NmOsek_NodeConfigType                                                                                                                                                                                                                                                      |

## Reference Documents

| No.  | Source  | Title                                           | Version      |
|------|---------|-------------------------------------------------|--------------|
| [1]  | AUTOSAR | AUTOSAR_SWS_CANNetworkManagement.pdf            | V3.3.0       |
| [2]  | AUTOSAR | AUTOSAR_SWS_DevelopmentErrorTracer.pdf          | V3.2.0       |
| [3]  | AUTOSAR | AUTOSAR_SWS_NetworkManagementInterface.pdf      | V3.0.0       |
| [4]  | AUTOSAR | AUTOSAR_TR_BSWModuleList.pdf                    | V1.6.0       |
| [5]  | OSEK    | <a href="#">nm253.pdf</a>                       | 2.5.3        |
| [6]  | Vector  | TechnicalReference_NmOsek_<OEM>.pdf             | see delivery |
| [7]  | Vector  | TechnicalReference_Nm.pdf                       | see delivery |
| [8]  | Vector  | TechnicalReference_CanSM.pdf                    | see delivery |
| [9]  | Vector  | TechnicalReference_Rte.pdf                      | see delivery |
| [10] | Vector  | Startup_<OEM>_SLP<SLP>.pdf                      | see delivery |
| [11] | Vector  | TechnicalReference_MICROSAR_IdentityManager.pdf | see delivery |
| [12] | Vector  | TechnicalReference_Com.pdf                      | see delivery |
| [13] | Vector  | TechnicalReference_CanIf.pdf                    | see delivery |

## Scope of the Document

This technical reference describes the general use of the OSEK Network Management basis software.



### Caution

We have configured the programs in accordance with your specifications in the questionnaire. Whereas the programs do support other configurations than the one specified in your questionnaire, Vector's release of the programs delivered to your company is expressly restricted to the configuration you have specified in the questionnaire.

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# 1 Introduction

This document describes the functionality, API and configuration of the MICROSAR BSW Complex Device Driver module NmOsek.

|                                          |                                                         |                                                       |
|------------------------------------------|---------------------------------------------------------|-------------------------------------------------------|
| <b>Supported AUTOSAR Release*:</b>       | 4                                                       |                                                       |
| <b>Supported Configuration Variants:</b> | pre-compile, post-build-loadable, post-build-selectable |                                                       |
| <b>Vendor ID:</b>                        | NMOSEK_VENDOR_ID                                        | 30 decimal<br>(= Vector-Informatik, according to HIS) |
| <b>Module ID:</b>                        | NMOSEK_MODULE_ID                                        | 255 decimal<br>(according to ref. [4])                |

\* For the precise AUTOSAR Release 4.x please see the release specific documentation.

This document describes the handling of the software component “MICROSAR OSEK Direct Network Management (Nm\_AsrNmDirOsek)”.

That means:

- > How to integrate and initialize
- > How to configure
- > How to operate
- > Basic understanding of the NmOsek component

Additionally, this document describes the API functions of NmOsek.

For details on the NM algorithms, please refer to the specification of the OSEK [5].

## 1.1 Architecture Overview

The following figure shows where the NmOsek is located in the AUTOSAR architecture.

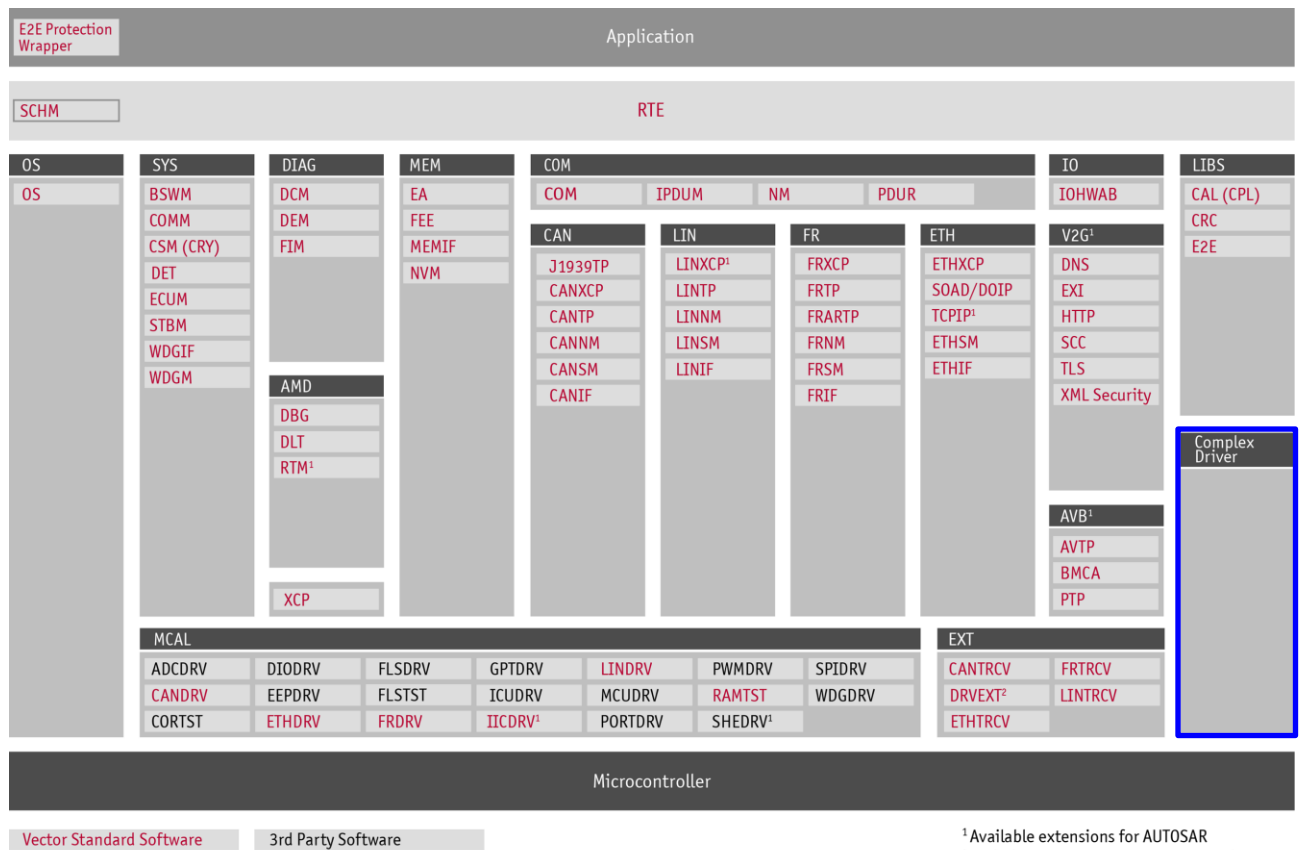


Figure 1-1 AUTOSAR 4.x Architecture Overview

The next figure shows the interfaces to adjacent modules of the NmOsek. These interfaces are described in chapter 4.

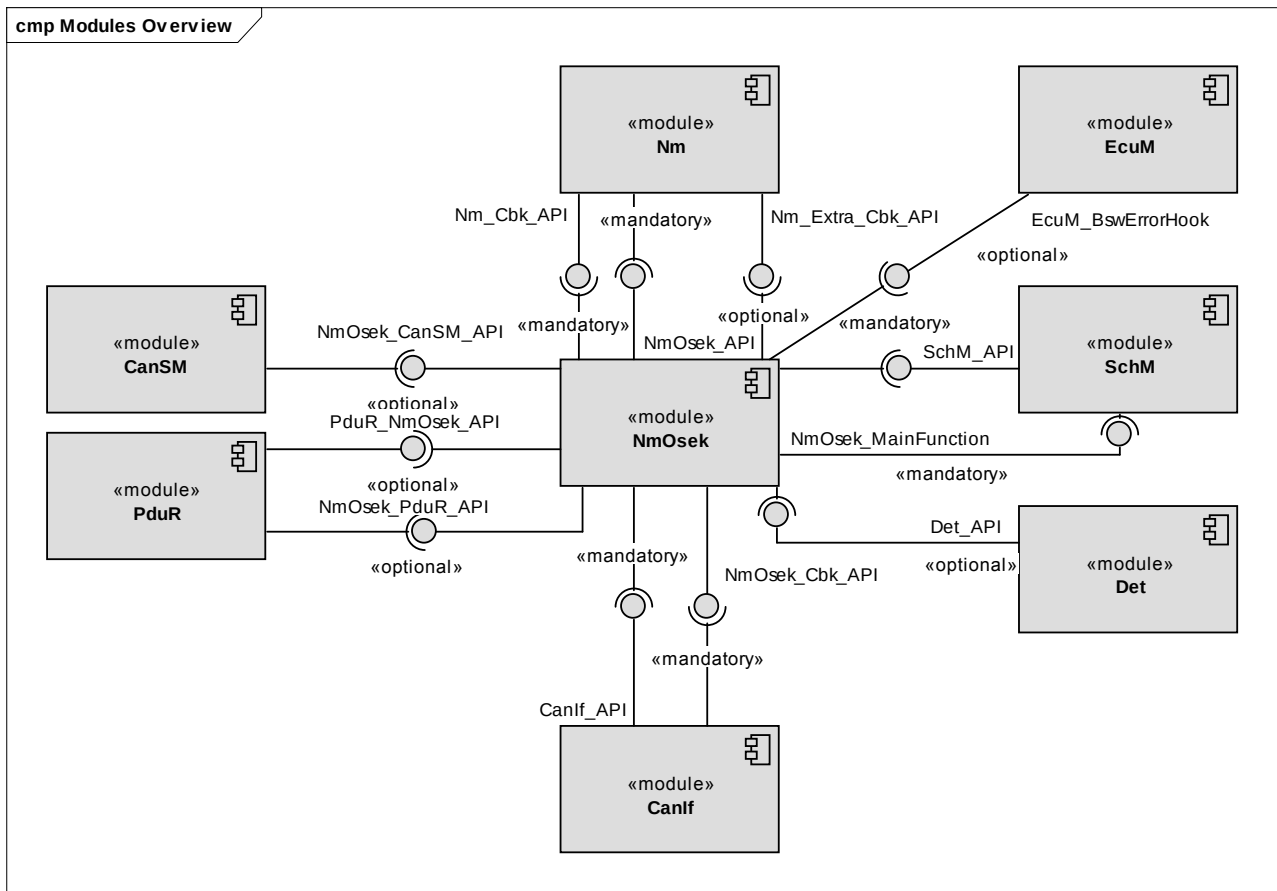


Figure 1-2 Interfaces to adjacent modules of the NmOsek

Note that NmOsek does not provide any service ports for the RTE so that applications can access services via the RTE. However there are some features that allow applications to access services offered by NmOsek. These features are described in the following chapter.

## 2 Functional Description

### 2.1 Features

The features listed in the following tables cover the complete functionality specified for the NmOsek.

Since NmOsek is a Complex Device Driver it is not part of the AUTOSAR standard. However some of its features are similar to those of the AUTOSAR CAN Network Management Basic Software Module whose functionality is specified in [1]. NmOsek is implemented as CAN-specific 'Generic BusNm' as suggested in [3]. Usually the standard AUTOSAR Network Management consists of the following components:

1. Network Management Interface (Nm)<sup>1</sup>
2. CAN Network Management (CanNm)<sup>1</sup>
3. FlexRay Network Management (FrNm)<sup>1</sup>
4. LIN Network Management (LinNm)<sup>1</sup>
5. UDP Network Management (UdpNm)<sup>1</sup>
6. J1939 Network Management (J1939Nm)<sup>1</sup>

The Network Management Interface (1) as specified in [3] may also be used in combination with a Generic BusNm instead of the standard BusNms (2-5) on each controller. For details about this concept please refer to [3].

A Generic BusNm needs to fulfill some AUTOSAR features. These are given in the following tables:

> Table 2-1 Supported AUTOSAR features

> Table 2-2 Not supported AUTOSAR features

For further information of not supported features see also chapter 5.

Vector Informatik provides further NmOsek functionality of the Direct OSEK Network Management that is specified in the OSEK Network Management specification [5].

> Table 2-3 Supported OSEK NM features

Since not all these features are applicable in an AUTOSAR environment there are some replacements which are listed in

> Table 2-4 OSEK NM replacement features.

The following features specified in [1] are supported:

| Supported AUTOSAR Features                                                                      |
|-------------------------------------------------------------------------------------------------|
| Operational Modes and Network States in analogy to the CAN Network Management (see section 2.3) |
| Initialization (see section 2.2)                                                                |
| Remote Sleep Indication (see section 2.10.4)                                                    |

<sup>1</sup> Not covered by this document

| Supported AUTOSAR Features                                      |
|-----------------------------------------------------------------|
| User Data Handling (see section 2.10.5)                         |
| Passive Mode (see section 2.10.1)                               |
| Node Identifier Services (see section 2.10.6)                   |
| Network Management PDU Rx Indication (see section 2.10.7)       |
| State Change Notification (see section 2.10.8)                  |
| Communication Control (see section 2.10.9)                      |
| Transmission Error Handling (always active) (see section 4.4.1) |
| Post-Build Loadable                                             |
| MICROSAR Identity Manager using Post-Build Selectable           |

Table 2-1 Supported AUTOSAR features

The following features specified in [1] are not supported:

| Not Supported AUTOSAR Features |
|--------------------------------|
| Bus Load Reduction Mechanism   |
| AUTOSAR Debugging Concept      |

Table 2-2 Not supported AUTOSAR features

The following features specified in [5] are also provided:

| OSEK NM Features                                                                   |
|------------------------------------------------------------------------------------|
| Providing Advanced Status Information (see section 2.10.3)                         |
| Controlling the transition to the bus sleep state and vice versa (see section 2.7) |
| Determination and monitoring of the network configuration (see section 2.10.2)     |

Table 2-3 Supported OSEK NM features

The following features are provided as a replacement for the OSEK features:

| Replacement OSEK NM Features                                                                             |
|----------------------------------------------------------------------------------------------------------|
| BusOff Occurrence and Recovery Notification (instead of BusOff handling by the module) (see section 2.9) |
| Exchanging user data the AUTOSAR way (see section 2.10.5)                                                |

Table 2-4 OSEK NM replacement features

## 2.2 Initialization

Before the NmOsek can be used it has to be initialized by the application. The initialization has to be carried out before any other functionality of NmOsek is executed. It shall take place after initialization of the CAN Interface and prior to the initialization of the Network Management Interface.

Also refer to the function definition of NmOsek\_Init() in section 4.2.1.1.



### Caution

NmOsek assumes some variables to be initialized to zero at start-up. If your embedded target does not initialize RAM within the start-up code the function 'NmOsek\_InitMemory()' has to be called during start-up and before the actual component initialization has been performed. Please also refer to the function definition of NmOsek\_InitMemory() in section 4.2.1.

## 2.3 States

There are several states and sub-states within the Direct OSEK NM:

- > NmOn/NmOff
- > NmAwake/NmWaitSleep/NmBusSleep
- > NmReset
- > NmNormal/NmLimpHome
- > NmRun/NmPrepSleep
- > NmActive/NmPassive

### 2.3.1 Overview

The mentioned states have a hierarchical linkage (see Figure 2-1).

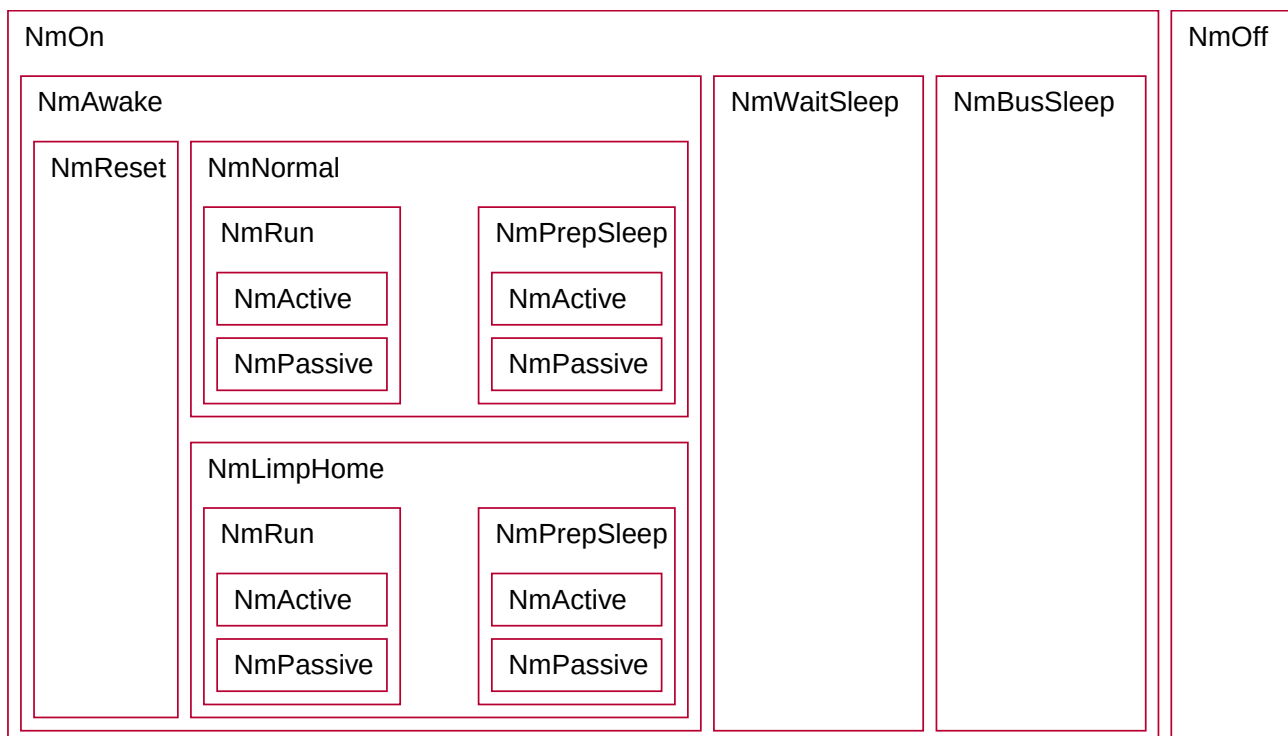


Figure 2-1 States of NmOsek in hierarchical context

| Reference name | AUTOSAR NM State                                                                               | Application status                                      | NM states                      | Bus    | CAN     |
|----------------|------------------------------------------------------------------------------------------------|---------------------------------------------------------|--------------------------------|--------|---------|
| Reset          | Repeat Message<br>Wait Network Startup <sup>2</sup><br>Wait Network Gw Msg Active <sup>2</sup> | The application may or may not require communication.   | NmOn<br>NmAwake<br>NmReset     | AWAKE  | Online  |
| Normal         | Repeat Message<br>Normal Operation<br>Wait Network Gw And Event Msg Active <sup>2</sup>        | The application requires communication.                 | NmOn<br>NmAwake<br>NmRun       | AWAKE  | Online  |
| SleepInd       | Repeat Message<br>Ready Sleep                                                                  | The application does not require communication anymore. | NmOn<br>NmAwake<br>NmPrepSleep | AWAKE  | Online  |
| WaitBusSleep   | Prepare Bus Sleep                                                                              | The application does not require communication.         | NmOn<br>NmWaitSleep            | AWAKE  | Online  |
| Sleep          | Bus Sleep                                                                                      | The application does not require communication.         | NmOn<br>NmBusSleep             | ASLEEP | Offline |
| PowerOff       | (uninitialized)                                                                                | N/A                                                     | N/A                            | ASLEEP | N/A     |

Table 2-5 States of OSEK NM

### 2.3.2 NmOn / NmOff

The state NmOn is the normal state of NmOsek: NmOsek is ready to take part in the logical ring. It sends its NM messages and monitors the network. Internal state transitions can occur. NmOn is entered after initialization (i.e. NmOsek\_Init() has been called).

NmOsek cannot be switched to the NmOff state at runtime. NmOsek remains disabled until NmOsek\_Init() was called. If it is disabled it does not take part in the logical ring. There are no NM messages from this node and no internal state transitions of NmOsek. Especially, there will be no transition to BusSleep.

### 2.3.3 NmAwake / NmWaitBusSleep / NmBusSleep

State NmWaitBusSleep is entered if all participating NM nodes have signaled their readiness for BusSleep, i.e. a NM message with a set SleepAcknowledge flag was received or transmitted.

This state is kept for some time before the transition to NmBusSleep occurs automatically.

State NmAwake is entered if there is any need for communication.

<sup>2</sup> Only available if 'Wake Up Frame Enabled' is turned ON in the configuration (setting might not be available in your delivery)



### 2.3.4 NmReset

NmReset exists in order to start building the logical ring. Therefore, when it is left, the ALIVE message is being sent.

NmReset is a transient state (a state that is entered and being left to the next state without an extra trigger) if it is entered when the logical ring needs to be re-established (e.g. due to a missing RING message reception, i.e. TMax timer has timed out or due to a reason to leave NmLimpHome / NmWaitBusSleep / NmBusSleep). The next state is NmNormal if the reception/transmission attempts do not exceed rx\_limit/tx\_limit (see [5] for details). Otherwise the next state is NmLimpHome.

NmReset may also be a non-transient state if 'Wake Up Frame Enabled' is turned ON in the configuration<sup>2</sup>. In this case NmReset is entered after a wake-up event has occurred (communication request by higher layer or wake-up event on CAN bus) in case TPWONWaitBusSleep<sup>2</sup> has elapsed. When it has been entered in this case, a timer is started with a value of TBWDisableSend<sup>2</sup>. OSEK NM then remains in NmReset until TBWDisableSend<sup>2</sup> has elapsed.

### 2.3.5 NmNormal / NmLimpHome

NmNormal is the default state if NmOsek is awake. In this state, NmOsek runs the algorithm to establish and run the logical ring. NmOsek can also determine the current network configuration.

State NmLimpHome is entered if a fatal bus error occurs or if NmOsek can't send or receive NM messages. In this state NmOsek cyclically tries to send so-called LimpHome messages. If message transmission and reception works again (e.g. bus error has been removed), this state is left.

### 2.3.6 NmRun / NmPrepSleep

If NmOsek is in state NmActive in the context of NmRun and NmNormal the node sends its NM message according to the specification.

NmPrepSleep is entered when NmOsek\_NetworkRelease() is called. This API indicates that this node does not need communication anymore and the NM node waits for the network to enter BusSleep mode.

NmRun is entered when NmOsek\_NetworkRequest() is called. Note that neither NmOsek\_NetworkRelease() nor NmOsek\_NetworkRequest() are available if the 'Passive Mode Enabled' switch is activated in the NM Interface configuration. In this case the state NmRun is also not available.

### 2.3.7 NmActive / NmPassive

The NmPassive state becomes available if the 'Passive Mode Enabled' feature is activated in the NM Interface configuration or the 'Com Control Enabled' feature is activated in the NM Interface configuration (see also section 2.10.9).

The NmActive state is unavailable if the 'Passive Mode Enabled' feature is enabled. So NmOsek is always in NmPassive in this case (refer to [7] for further details). That implies that the application can never request communication.

NmOsek is normally in NmActive if the 'Passive Mode Enabled' feature is disabled. In that case a transition to NmPassive is only possible if the 'Com Control Enabled' feature is activated in the Network Management Interface configuration. Table 2-6 provides a summary

of these configuration possibilities. Note that 'Passive Mode Enabled' and 'Com Control Enabled' cannot be active at the same time.

The state NmActive is the normal state of NmOsek: NmOsek actively takes part in the logical ring. In case the 'Com Control Enabled' feature is activated, NmActive can be activated (if not already active) by calling API NmOsek\_EnableCommunication().

For the NmPassive state some limitations have to be considered. This state prevents NmOsek from active participation in the ring. That implies that the own network node cannot keep that network awake. In case the 'Com Control Enabled' feature is activated and the current NmOsek state is NmAwake (i.e. neither NmBusSleep nor NmWaitBusSleep), NmPassive can be entered by calling the API service NmOsek\_DisableCommunication(). Note that the communication must not be released by the application until NmActive has been entered again. If an ECU is in NmPassive state, it remains in its previous main state. It listens to the bus without sending any own NM message.

| NM Interface Configuration |  | Passive Mode Enabled: Off<br>Com Control Enabled: Off |   | Passive Mode Enabled: Off<br>Com Control Enabled: On |  | Passive Mode Enabled: On<br>Com Control Enabled: Off |  | Passive Mode Enabled: On<br>Com Control Enabled: On |     |
|----------------------------|--|-------------------------------------------------------|---|------------------------------------------------------|--|------------------------------------------------------|--|-----------------------------------------------------|-----|
| State Availability         |  |                                                       |   |                                                      |  |                                                      |  |                                                     |     |
| NmActive                   |  | ■                                                     | ■ |                                                      |  |                                                      |  |                                                     |     |
| NmPassive                  |  |                                                       | ■ | ■                                                    |  |                                                      |  |                                                     | N/A |

Table 2-6 Availability of NmActive / NmPassive in dependency of Nm Configuration Settings

Please take a look at [1] and [3] for further details about the 'Communication Control' feature. There is no token inspection and there are no state transitions. However, application messages can still be sent.

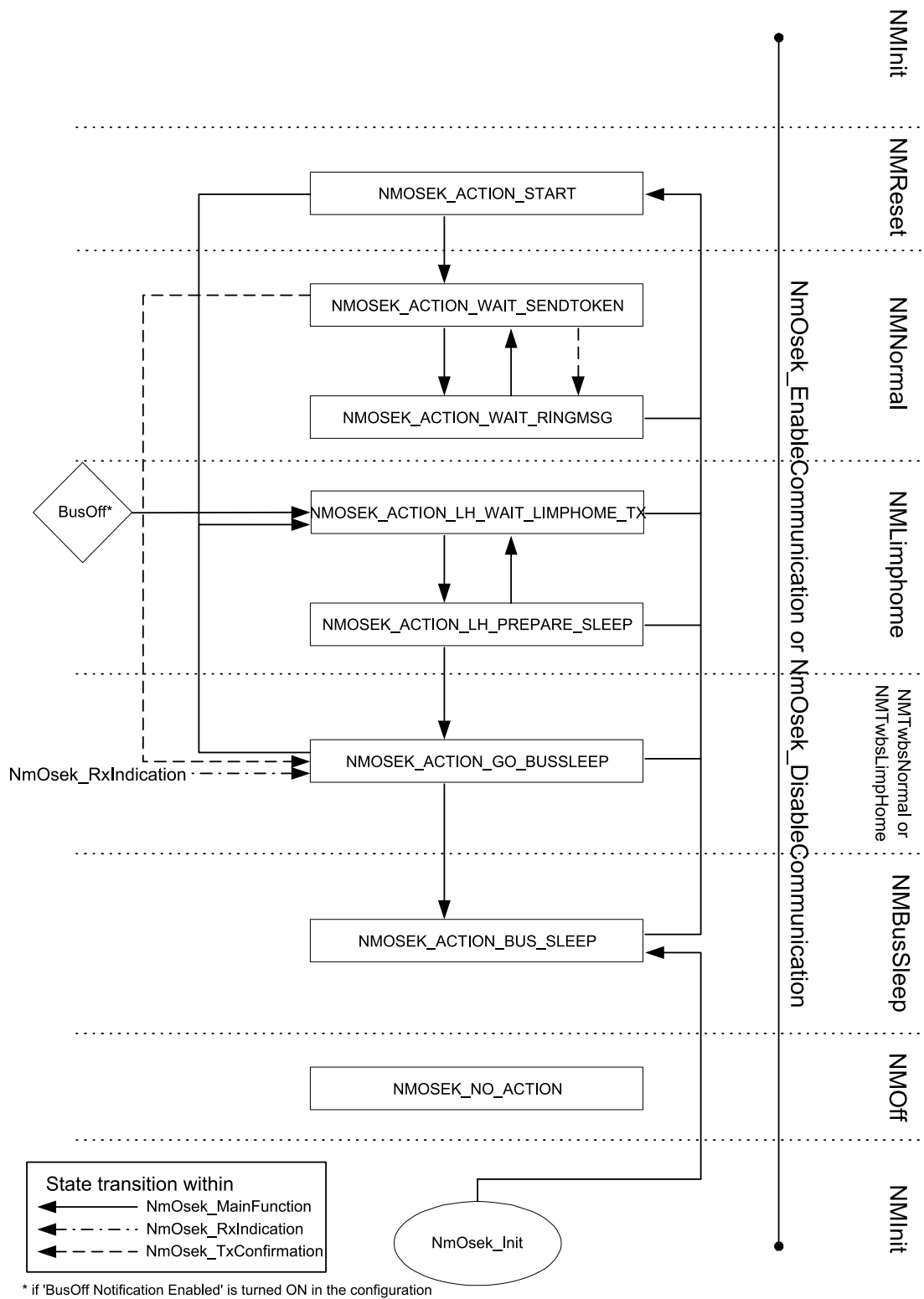


Figure 2-2 Internal state-machine of NmOsek

## 2.4 Main Function

There is only one main function provided by NmOsek called 'NmOsek\_MainFunction'. This function needs to be called cyclically in the interval configured in the main configuration

settings (refer to section 4.2.1.18 for details). This is usually the task of the AUTOSAR Schedule Manager (SCHM) being part of the RTE. Please refer to [9] for further details about configuring main functions.

## 2.5 Error Handling

### 2.5.1 Development Error Reporting

By default, development errors are reported to the DET using the service `Det_ReportError()` as specified in [2], if development error reporting is enabled (i.e. pre-compile parameter `NMOSEK_DEV_ERROR_REPORT==STD_ON`).

If another module is used for development error reporting, the function prototype for reporting the error can be configured by the integrator, but must have the same signature as the service `Det_ReportError()`.

The reported NmOsek ID is 255. The instance ID of NmOsek can be configured at pre-compile time.

The reported service IDs identify the services which are described in 4.2. The following table presents the service IDs and the related services. If a service is not listed in this table, refer also to [6]:

| Service ID | Service                         |
|------------|---------------------------------|
| 0x00       | NmOsek_Init()                   |
| 0x01       | NmOsek_PassiveStartUp()         |
| 0x02       | NmOsek_NetworkRequest()         |
| 0x03       | NmOsek_NetworkRelease()         |
| 0x04       | NmOsek_SetUserData()            |
| 0x05       | NmOsek_GetUserData()            |
| 0x06       | NmOsek_GetNodeIdentifier()      |
| 0x07       | NmOsek_GetLocalNodeIdentifier() |
| 0x08       | NmOsek_RepeatMessageRequest()   |
| 0x0A       | NmOsek_GetPduData()             |
| 0x0B       | NmOsek_GetState()               |
| 0x0C       | NmOsek_DisableCommunication()   |
| 0x0D       | NmOsek_EnableCommunication()    |
| 0x0F       | NmOsek_TxConfirmation()         |
| 0x10       | NmOsek_RxIndication()           |
| 0x13       | NmOsek_MainFunction()           |
| 0x15       | NmOsek_Transmit()               |
| 0x20       | NmOsek_GetStatus()              |
| 0x21       | NmOsek_GetConfig()              |
| 0x22       | NmOsek_CmpConfig()              |
| 0x30       | NmOsek_CanSM_BusOffBegin()      |
| 0x31       | NmOsek_CanSM_BusOffEnd()        |

| Service ID | Service                             |
|------------|-------------------------------------|
| 0xC0       | NmOsek_RequestBusSynchronization()  |
| 0xD0       | NmOsek_CheckRemoteSleepIndication() |
| 0xE0       | NmOsek_CancelWaitForTxConfOrRxInd() |
| 0xF1       | NmOsek_GetVersionInfo()             |

Table 2-7 Service IDs

The errors reported to DET are described in the following table:

| Error Code | Description                                                                                                                                                                   |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0x01       | NMOSEK_E_UNINIT<br>API service used without module initialization or service called for inactive channel in this pre-defined variant if Post-Build-Selectable support is used |
| 0x02       | NMOSEK_E_INVALID_CHANNEL<br>API service used with wrong channel handle                                                                                                        |
| 0x03       | NMOSEK_E_INIT_FAILED<br>NmOsek initialization has failed                                                                                                                      |
| 0x04       | NMOSEK_E_WRONG_WAIT_BUS_SLEEP_FLAG<br>WaitBusSleep Flag is not correctly set                                                                                                  |
| 0x05       | NMOSEK_E_WRONG_BUS_SLEEP_FLAG<br>BusSleep Flag is not correctly set                                                                                                           |
| 0x0A       | NMOSEK_E_PARAM_CONFIG<br>API service NmOsek_Init() called with wrong parameter                                                                                                |
| 0x0C       | NMOSEK_E_PARAM_POINTER<br>API service used with invalid pointer parameter (NULL)                                                                                              |
| 0x11       | NMOSEK_E_ALREADY_INITIALIZED<br>The service NmOsek_Init() is called while the module is already initialized                                                                   |
| 0x21       | NMOSEK_E_PDUR_TRIGGER_TX_ERROR<br>Call of PduR_NmOsekTriggerTransmit() failed due to an unexpected PDU length                                                                 |
| 0x33       | NMOSEK_E_INVALID_GENDATA<br>The value of a variable used for write accesses is out of bounds                                                                                  |

Table 2-8 Errors reported to DET

## 2.5.2 Production Code Error Reporting

Currently the NmOsek module does not define any error that can be reported to the DEM module.

## 2.6 Standard Use Cases

The following table provides some standard use cases how services of the NmOsek are invoked.

| Reference                                             | Description                                                                                                                                                                                                                                                                                                                                                                    | associated NmOsek services |
|-------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| Start (active) network operation                      | <p>The system requires CAN communication and starts NmOsek.</p> <p>The system is not ready to sleep.</p> <p>NmOsek is informed that it requires the bus.</p> <p>If the current NM state is not NmAwake the function Nm_NetworkMode() is called (refer to [7] for details) within the next call of NmOsek_MainFunction().</p>                                                   | NmOsek_NetworkRequest()    |
| Start (passive) network operation                     | <p>It is required to provide CAN communication due to bus activity. NmOsek is started passively.</p> <p>The system itself is ready to sleep. NmOsek is informed that it does not require the bus actively.</p> <p>If the current NM state is not NmAwake the function Nm_NetworkMode() is called (refer to [7] for details) within the next call of NmOsek_MainFunction().</p> | NmOsek_PassiveStartUp()    |
| Initialization                                        | <p>NmOsek is initialized in SLEEP mode.</p> <p>The communication path at that time is switched off.</p>                                                                                                                                                                                                                                                                        | NmOsek_Init()              |
| Set sleep indication                                  | <p>The system is ready to sleep and does not need the bus anymore.</p>                                                                                                                                                                                                                                                                                                         | NmOsek_NetworkRelease()    |
| Cancel sleep indication                               | <p>The system is no more ready to sleep and needs the bus again.</p>                                                                                                                                                                                                                                                                                                           | NmOsek_NetworkRequest()    |
| Stop the network operation                            | <p>All ECUs in the network are ready to sleep.</p> <p>NmOsek has set itself into the BusSleep mode.</p> <p>NmOsek announces the successful transition to BusSleep mode with the help of Nm_BusSleepMode() (preceded by a call of Nm_PrepareBusSleepMode() when NmWaitBusSleep has been entered, details see [7]).</p>                                                          | —                          |
| External wake-up caused by starting network operation | <p>The system operates in local mode.</p> <p>An external wake-up event occurs.</p> <p>The system detects that communication is required.</p> <p>As the system requires the bus itself, NmOsek is started without a set SleepIndication flag.</p>                                                                                                                               | NmOsek_NetworkRequest()    |
| Demand network operation by the CAN bus               | <p>The system operates in local mode. The bus is asleep.</p> <p>The CAN controller detects a dominant level on the CAN bus.</p> <p>The NM node has to participate in the bus traffic because it was woken up by the bus.</p> <p>As the system does not require communication itself, NmOsek is started with a set SleepIndication flag.</p>                                    | NmOsek_PassiveStartUp()    |

|                                                               |                                                                                                                                                                                                            |               |
|---------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| Local wake-up caused by an event without communication demand | The system operates in local mode.<br>An external wake-up event occurs. This event can be processed locally. There is no need to activate the communication. Therefore NmOsek is started in BusSleep mode. | NmOsek_Init() |
| switch off the CPU (Power-Off)                                | The system switches off the CPU with the help of a „power on logic“.                                                                                                                                       | —             |

Table 2-9 Standard use cases

## 2.7 Control the Transition to the BusSleep Mode

NmOsek ensures that all active ECUs within a network will enter the BusSleep mode simultaneously.

This is required because an ECU will be woken up by any message on the CAN bus. That is the reason why a transition of a single ECU into the BusSleep mode only makes sense, if all the other ECUs are also ready for BusSleep mode and will not send further messages.

The NM algorithm (see [5]) detects the need to enter BusSleep mode or to wake up the bus again. That means that NmOsek decides when the CAN channel must be activated or deactivated.

NmOsek is ready for BusSleep if the system decides that no more CAN communication is needed. In this case NmOsek\_NetworkRelease() was called by the system. NmOsek is also ready for BusSleep if only NmOsek\_PassiveStartUp() was called but not NmOsek\_NetworkRequest(). The readiness for BusSleep remains until the NmOsek service NmOsek\_NetworkRequest() is called.

The transition to BusSleep may start when all network nodes are ready to sleep, i.e. Ring messages by all participants in the logical ring have been sent where the Sleep Indication bit is set. The last Ring message before the transition to BusSleep starts has both the Sleep Indication and the Sleep Acknowledge bit set. The next state is NmWaitBusSleep and Nm\_PreparesBusSleepMode() is called when this state has been entered. The transition from NmWaitBusSleep to NmBusSleep takes place if there is no wake-up event after NmWaitBusSleep has been entered.

NmOsek calls the function Nm\_BusSleepMode() after the transition into the BusSleep mode. The successful transition to BusSleep mode can also be checked by the status flag 'BusSleep' of NmOsek (see section 2.10.3) or the NmOsek\_GetState() function (see section 4.2.1.14).

## 2.8 Network Management Message Transmission and Reception

### 2.8.1 AUTOSAR CAN Interface

NmOsek requests the transmission of NM messages by calling the service CanIf\_Transmit. The application has to take care of the user data. For details refer to section 2.10.5.

The successful transmission of every network management message is confirmed by the CAN Interface with the service

```
void NmOsek_TxConfirmation(PduIdType nmOsekNmTxPduId)
```

The CAN Interface indicates the reception of NM message by calling the service

```
void NmOsek_RxIndication(PduIdType nmOsekRxPduId,
    const uint8* CanSduPtr, const Can_IdType canId)
```

Hint: these functions are called by NmOsek\_TxConfirmation\_X() / NmOsek\_RxIndication\_X(), X = 0,...,(Number of configured OSEK NM channels - 1) .

## 2.8.2 PDU Message Layout

The default PDU Message Layout is described in the following table:

|        | Bit 7                  | Bit 6 | Bit 5 | Bit 4 | Bit 3 | Bit 2 | Bit 1 | Bit 0 |
|--------|------------------------|-------|-------|-------|-------|-------|-------|-------|
| Byte 7 | User data 5 / Unused   |       |       |       |       |       |       |       |
| Byte 6 | User data 4 / Unused   |       |       |       |       |       |       |       |
| Byte 5 | User data 3 / Unused   |       |       |       |       |       |       |       |
| Byte 4 | User data 2 / Unused   |       |       |       |       |       |       |       |
| Byte 3 | User data 1 / Unused   |       |       |       |       |       |       |       |
| Byte 2 | User data 0 / Unused   |       |       |       |       |       |       |       |
| Byte 1 | Operation Code         |       |       |       |       |       |       |       |
| Byte 0 | Destination Identifier |       |       |       |       |       |       |       |

Table 2-10 PDU NM Message Layout

The NM PDU elements are given in Table 2-11. For details about these bytes please refer to the OSEK specification [5] and to the OEM-specific Technical Reference [6].

| NM PDU Element         | Description                                                                                                                                                                                                                                                                                           |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Destination Identifier | Recipient node identifier / successor in the logical ring that shall send the next NM Message                                                                                                                                                                                                         |
| Operation Code         | Operation Code that contains the NM message type (ALIVE, RING, LIMPHOME message) and Sleep Flags (Sleep.ind, Sleep.ack).                                                                                                                                                                              |
| User Data              | Optional user data bytes that might be filled in the NM TX PDU by using the Nm API Nm_SetUserData() and may be retrieved by calling Nm_GetUserData() (see also [7]).<br>As an alternative, User Data may also be set by using Com Signals (if 'Com User Data Support' is turned ON in the Nm module). |

Table 2-11 NM PDU Elements

ALIVE and RING messages are sent in NmNormal and NmNormalPrepSleep, LIMPHOME messages are sent in NmLimpHome. For more details, please refer to [5].

The NM message contains a Destination Identifier byte. This byte refers to the logical successor in the ring. The node addresses itself, if there is currently no logical successor.

For the algorithm how the Destination Identifier byte/logical successor is determined refer to chapter 2.2.6 of [5].

Furthermore, the NM Message contains a Sleep Indication bit and a Sleep Acknowledge bit.

The Sleep Indication bit is set to 1 if the own node is ready to sleep, otherwise to 0.

The Sleep Acknowledge bit is only set in RING messages. If all nodes have indicated their readiness to sleep during one ring cycle, the next node in the ring will send a RING message



with Sleep Acknowledge bit set to 1. In other words, a node sends the Sleep Acknowledge Bit set to 1 if no NM message with the Sleep Indication bit set to 0 has been received during the last ring cycle. The Sleep Acknowledge bit is never set in ALIVE or LIMPHOME messages.

The last LIMPHOME message from a node before it goes to sleep has the Sleep Indication bit set to 1, the Sleep Acknowledge bit set to 0.

### 2.8.3 TX Message Observation

NmOsek provides the possibility to observe OSEK NM message transmission requests. This can be achieved by two possible approaches:

- > defining the 'Tx Confirmation Timeout Time' parameter in the NmOsek channel configuration
- > using the 'Tx Deadline Monitoring' mechanism of Com.

The basic algorithm of the first approach works as follows, the second approach is described in chapter 2.10.12:

- > OSEK NM issues a transmission request for a NM message (CanIf\_Transmit()).
- > If the transmission request has been accepted (CanIf\_Transmit() returns E\_OK), the Message Timeout Timer is started with the configured 'Tx Confirmation Timeout Time'.
- > If a message transmission is successful on the bus, the Message Timeout Timer is stopped within NmOsek\_TxConfirmation().
- > If the Message Timeout Timer expires, Nm\_TxTimeoutException() is called.
- > If another NM message transmit request is accepted, the Message Timeout Timer is restarted with the configured 'Tx Confirmation Timeout Time'.

This feature is mandatory for the feature 'First Message shall be NM Message' (see also chapter 2.10.11) and may also be mandatory for other OEM-specific features.

## 2.9 BusOff Occurrence and Recovery Notification

In AUTOSAR environments, CANSM is usually responsible for BusOff recoveries. Therefore, in contradiction to [5], OSEK NM is not responsible for BusOff recoveries but can be notified about BusOff occurrences by CanSM.

When a BusOff occurs the transition to NmLimpHome takes place if the CANSM notifies NmOsek about a BusOff occurrence. The timer TError is started. NmLimpHome may be left when the CANSM notifies NmOsek about a BusOff recovery and the timer TError has elapsed.

In order to use this feature, please make sure that the 'BusOff Notification Enabled' feature is activated in the NmOsek settings in the configuration tool. Also verify that the 'BusOff Begin / End' notifications are as well activated in the CANSM settings.

**Caution**

It is highly recommended to turn the 'BusOff Notification Enabled' setting ON in the configuration tool to have an OSEK NM behavior that conforms to [5].

If the 'BusOff Notification Enabled' feature is activated in the NmOsek configuration the implementation of NmOsek\_CanSM\_BusOffBegin() and NmOsek\_CanSM\_BusOffEnd() is available. The call of these functions is considered in the next NmOsek\_MainFunction() function call (also refer to section 4.4.2 and 4.4.2.2 as well as the CANSM Technical Reference [8]).

As follows, the behavior of OSEK NM is described for two scenarios: either OSEK NM does not know about BusOff ('BusOff Notification Enabled' is turned OFF) or OSEK NM is notified by CANSM about the BusOff state ('BusOff Notification Enabled' is turned ON). Usually, the decision whether OSEK NM is notified about BusOff or not is decided by the OEM.

### 2.9.1.1 Behavior without Knowledge about BusOff State

In this setup, OSEK NM is not notified about BusOff occurrences and thus the transition to NmLimpHome may be delayed or even may not take place at all. This setup requires the 'BusOff Notification Enabled' setting to be disabled.

In case a BusOff occurs and the bus disturbance remains for a longer time (multiple BusOff events may be consequence), the transition to NmLimpHome occurs in case the number of missing TX Confirmations  $tx_{conf}$  and/or the number of RX Indications  $rx_{ind}$  for NM messages exceed their limit(s), that is if

$$(tx_{conf} > TxLimit) \vee (rx_{ind} > RxLimit)$$

evaluates to true.

Depending on the 'Main Function Period'  $t_{NmCycle}$  (unit: ms) and the limits  $TxLimit$  (default value: 8 times) and  $RxLimit$  (default value: 4 times) it takes at least

$$LH_{delayed}(TxLimit, RxLimit, T_{RingMax}, T_{RingTyp}, t_{NmCycle}) :=$$

$$\min(\lfloor TxLimit/2 \rfloor * (T_{RingMax} + T_{RingTyp}), RxLimit * (T_{RingMax} + T_{RingTyp})) + t_{NmCycle} + t_{Additional}$$

milliseconds until the transition to NmLimpHome transition takes place where  $t_{Additional}$  denotes the time it takes until the timer  $T_{RingTyp}$  or  $T_{RingMax}$  times out.

In the following figures, situations are depicted where a BusOff event occurs and the bus disturbance remains as long as it takes to enter NmLimpHome. Usually, NmLimpHome would be entered directly if OSEK NM had knowledge about the BusOff event and if BusOff has just been detected. Due to the lack of knowledge about BusOff, NmLimpHome is only entered due to the exceeded threshold of the  $TxLimit$  and/or  $RxLimit$ .

If BusOff occurs while another node is addressed by the last RING message, the sequence contains the attempts by OSEK NM to re-establish the RING by trying to send ALIVE and RING messages itself:

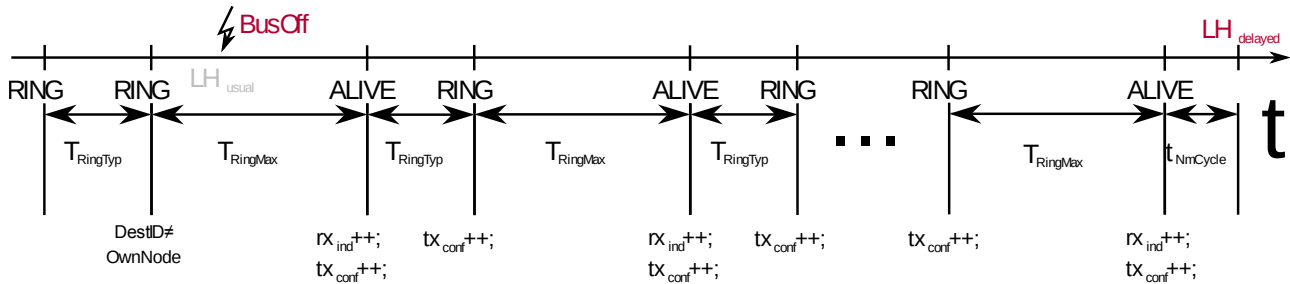


Figure 2-3 Temporal sequence when OSEK NM is not explicitly notified about the BusOff event if it is not addressed by the last RING message

The actual transition in Figure 2-3 is taken when the last increment of the  $tx_{conf}$  variable reaches  $TxLimit$ . This value is checked within the next main function call of `NmOsek` after  $t_{NmCycle}$  milliseconds. This point is indicated by  $LH_{delayed}$ . Using the default values and  $t_{NmCycle} := 20$  ms, the `NmLimpHome` transition offset calculates to  $LH_{delayed}(8, 4, 260, 100, 20) = 1460$  ms.

If BusOff occurs while the own node is addressed by the last RING message, the sequence of ALIVE and RING message looks like this:

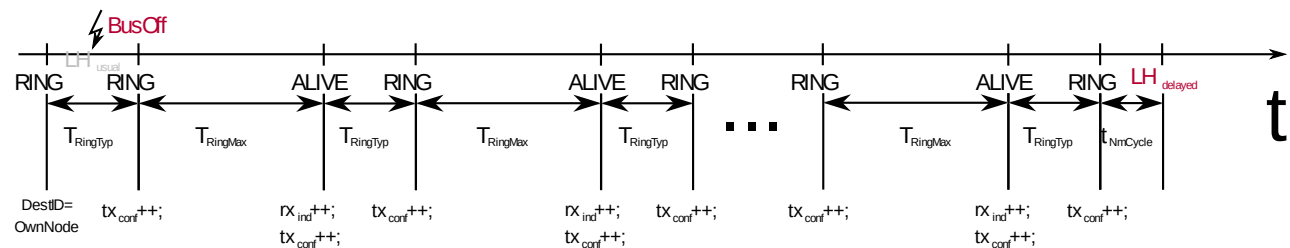


Figure 2-4 Temporal sequence when OSEK NM is not explicitly notified about the BusOff event if it is addressed by the last RING message

Note that there are other situations where BusOff may occur which are not illustrated here. The LimpHome messages are delayed by  $LH_{delayed}$ , too.



#### Note

As it can be seen, the missing knowledge about BusOff is a deviation to the original OSEK NM Specification [5].

### 2.9.1.2 Behavior with Knowledge about BusOff State

If OSEK NM is notified about the BusOff state by CANSM or another module (e.g. CDD), the behavior concerning the transition to `NmLimpHome` is in accordance with the original OSEK NM Specification [5]. This setup requires the 'BusOff Notification Enabled' setting to be enabled.

However, OSEK NM does not perform the BusOff recovery itself but waits for the BusOff End notification until the LimpHome message is tried to be sent.

So the timings concerning BusOff recoveries in CANSM need to be configured in accordance with the OSEK NM timings.

An example for the BusOff recovery by CANSM and the BusOff notification to OSEK NM is depicted in Figure 2-5.

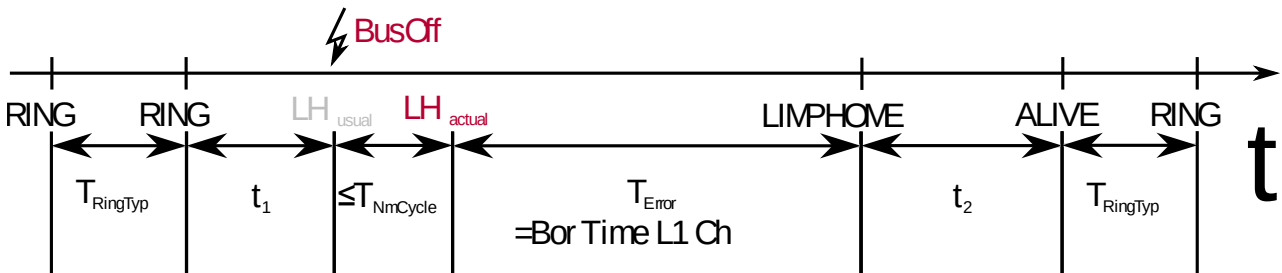


Figure 2-5 Temporal sequence when OSEK NM is notified about the BusOff event

In this example, after the last RING message has been received or transmitted, BusOff is detected after  $t_1$  and the NM is notified about BusOff occurrence (through NmOsek\_CanSM\_BusOffBegin). Then, NmLimpHome is entered at  $LH_{actual}$  in the next NmTask function call (thus the maximum delay is  $T_{NmCycle}$ ) and the timer  $T_{Error}$  is started. After  $T_{Error}$  has elapsed and the NM has been notified about a BusOff Recovery attempt (through NmOsek\_CanSM\_BusOffEnd), the NM LIMPHOME message is tried to be sent. If another NM message is received (in this example after  $t_2$ ) and the NM message has been sent successfully, the NM restarts with an NM ALIVE and an NM RING message.

In Table 2-12, the configuration settings of OSEK NM and CANSM are listed that need to be in accordance with each other.

| OSEK NM Setting             | CANSM Setting                                  |
|-----------------------------|------------------------------------------------|
| Module-wide settings        |                                                |
| BusOff Notification Enabled | BusOff Begin and BusOff End need to be defined |
| Channel-specific settings   |                                                |
| Limp Home Time              | Bor Time L2 Ch                                 |

Table 2-12 Settings of OSEK NM and CANSM to be configured in accordance if 'BusOff Notification' is used

CANSM settings of 'Bor Time L1 Ch' and 'Bor Time L2 Ch' being different are explicitly allowed.

Note that CANSM has different strategies for returning to the 'Bor Time L1 Ch' time if 'Bor Time L2 Ch' has been previously used as BusOff recovery time when the bus disturbance is gone.

## 2.10 Other Features

This section describes further features of the NmOsek module.

### 2.10.1 Passive Mode

Nodes in passive mode cannot transmit NM messages and therefore they do not actively participate in the network. Due to this fact passive nodes cannot request the network and the state changes are performed according to the last received NM message or timeouts.

This mode can be used for nodes that do not need to keep the bus awake to save resources.

NmOsek always stays in the NmPassive state. Please refer to section 2.3.7 for more details about the NmPassive state.

### 2.10.2 Determine and Monitor the Network Configuration

The network configuration is determined in the so-called “Start Phase”. Each NM node regards itself as ready for operation.

After the “Start Phase”, NmOsek monitors the network. This “Monitoring Phase” is used to detect new nodes as well as the loss of existing nodes.

This feature called ‘Use Nm Node List’ can be enabled/disabled in the configuration tool. In order to save runtime and memory resources, do not enable this feature unless you need it.

Please make sure that the ‘Number of Nm Config Nodes’ setting contains the maximum number of NM nodes in all networks (channels), i.e. check all channel-specific settings for the ‘NM Message Count’ attribute and verify that the ‘Number of Nm Config Nodes’ contains the maximum of all ‘NM Message Count’ values.

#### 2.10.2.1 Configuration Update by NmOsek

The network nodes that are currently active on the channel are stored inside a byte array. Each bit in this array represents a network node. A bit inside this array is set if a network node has been detected as present in the logical ring. The term ‘network configuration’ refers to this byte array.

The network configuration of NmOsek is updated each time a NM message is received, i.e. usually in ISR-context. Please be aware of this behavior if multiple decisions in the application are necessary to gain a result (e.g. multiple nested ‘if’). Maybe it is necessary to copy the current network configuration to a temporary variable.

The configuration is discarded when the ring is not stable anymore, e.g. if a node gets lost or a bus error occurs.

#### 2.10.2.2 Retrieve Configuration

The information about the network nodes that are currently active may be retrieved by using the API NmOsek\_GetConfig().

#### 2.10.2.3 Compare Configuration

NmOsek provides the possibility to compare the current network configuration with a reference configuration (see API NmOsek\_CmpConfig()).

This reference configuration is stored in two arrays that have to be provided by the application:

| Variable Name            | Description                                                                                                                                                                                                                  |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NmOsek_TargetConfigTable | Indicates the required status of the NM nodes. If the corresponding bit is set (‘1’), the associated NM node has to be present. If the bit is not set (‘0’), the node may not be present.                                    |
| NmOsek_ConfigMaskTable   | Indicates which NM nodes have to be checked. If the corresponding bit is set (‘1’), the presence of the associated NM node has to be evaluated. If the bit is not set (‘0’), the node is not of interest for the comparison. |

| Variable Name | Description                                                                                                |
|---------------|------------------------------------------------------------------------------------------------------------|
|               | The assignment between the node address and the corresponding bit/byte position is depicted in Table 2-14. |

Table 2-13 Reference Configuration Tables

| Byte | Bit7 | Bit6 | Bit5 | Bit4 | Bit3 | Bit2 | Bit1 | Bit0 |
|------|------|------|------|------|------|------|------|------|
| 0    | 7    | 6    | 5    | 4    | 3    | 2    | 1    | 0    |
| 1    | 15   | 14   | 13   | 12   | 11   | 10   | 9    | 8    |
| 2    | 23   | 22   | 21   | 20   | 19   | 18   | 17   | 16   |
| ...  | ...  | ...  | ...  | ...  | ...  | ...  | ...  | ...  |
| n/8  | n-1  | n-2  | n-3  | n-4  | n-5  | n-6  | n-7  | n-8  |

Table 2-14 Assignment between node address and bit/byte position for n stations

Bit 0 of byte 0 represents the NM node with station address 0. Bit 7 of the most significant byte represents the NM node with the highest station address.

An ECU which should be present in the reference configuration is marked with '1' at the corresponding position in both tables.

An ECU which should not be present in the reference configuration is marked with '1' at the corresponding position in NmOsek\_ConfigMaskTable and '0' in the NmOsek\_TargetMaskTable.

An ECU which should not be part of the comparison is marked with '0' at the corresponding position in NmOsek\_ConfigMaskTable.

NmOsek\_CmpConfig() compares the current configuration with the reference only for those nodes that are marked with '1' in ConfigMaskTable.

**Example**

```
#include "NmOsek.h"

/* ROM tables provided by the application */
CONST(NmOsek_NodeConfigType, NMOSSEK_APPL_CONST)
NmOsek_TargetConfigTable[2] =
{
    { 0x69, /* NM node 6,5,3,0 */ /* Configuration: Net1
    */
      0x00,
      0x80, /* NM node 23 */
      0x80 /* NM node 31 */
    },
    { 0x02, /* NM node 1 */ /* Configuration: LimpHome
    */
      0x12, /* NM node 12,9 */
      0x11, /* NM node 20,16 */
      0x40 /* NM node 30 */
    },
}
CONST(NmOsek_NodeConfigType, NMOSSEK_APPL_CONST)
NmOsek_ConfigMaskTable[2] =
{
    { 0xF9, /* NM node 7..3,0 */ /* Configuration: Net1
    */
      0x00,
      0x80, /* NM node 23 */
      0x80 /* NM node 31 */
    },
    { 0x0F, /* NM node 3..0 */ /* Configuration:
    LimpHome */
      0x12, /* NM node 12,9 */
      0x11, /* NM node 20,16 */
      0xC1 /* NM node 31,30,24 */
    },
}
/* application */
...
#define LIMPHOME_DESIREDCONFIG 1
...
if( NmOsek_CmpConfig(LIMPHOME_DESIREDCONFIG)==1 )
{...;}
```

**2.10.3 Provide Status Information**

NmOsek provides information on the status of the NM node and the network itself.

This status information can be accessed with the help of APIs NmOsek\_GetStatus(). These two services differ in the way the result is passed back. The result itself is of type uint8 (see Table 2-15).

| Bit | Name               | Interpretation                                                                                                                                                                                                            | Mask for bitwise AND |
|-----|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| 0   | Ring Stable        | Indicates if the current network is stable. Stable means that the configuration has not changed during the last ring cycle.<br>0 ring is not stable<br>1 ring is stable                                                   | 0x01                 |
| 1   | Bus Off Error      | Indicates if a Bus Off has occurred<br>0 no Bus Off error<br>1 Bus Off error occurred.<br>CAN is locked. CAN driver is offline.                                                                                           | 0x02                 |
| 2   | Active             | Indicates the operation mode of NmOsek<br>0 NMPassive (see chapter 2.3.7)<br>1 NMActive                                                                                                                                   | 0x04                 |
| 3   | LimpHome           | Indicates if NmOsek is in the mode LimpHome. LimpHome is entered, if NmOsek can't send or receive NM messages.<br>0 not in state LimpHome<br>1 in state LimpHome                                                          | 0x08                 |
| 4   | BusSleep           | Indicates if the bus is asleep or not<br>0 not in state BusSleep<br>1 in state BusSleep                                                                                                                                   | 0x10                 |
| 5   | WaitBusSleep       | Indicates if NmOsek currently waits for BusSleep<br>0 NmOsek is not in state NmWaitBusSleep<br>1 NmOsek is in state NmWaitBusSleep                                                                                        | 0x20                 |
| 6   | TxRingData Allowed | Indicates if a write access on the user ring data is allowed or not.<br>0 write access on ring data is allowed<br>1 write access on ring data is not allowed<br>Note: This flag is not supported by all OEMs <sup>3</sup> | 0x40                 |
| 7   | SleepIndication    | Indicates if the application has signaled readiness for Sleep or not.<br>0 GotoMode(Awake) was called<br>1 GotoMode(SleepInd) was called                                                                                  | 0x80                 |

Table 2-15 Access on NmOsek status information

The OSEK specification [5] mentions a status flag that indicates if NmOsek is started or not (NmOn, NmOff). This status flag is not supported by the NmOsek component. Also refer to the OSEK specification for further details about these status flags.

The status information can be evaluated as follows:

<sup>3</sup> This flag is only set to 0 if the 'Token Monitoring' feature is enabled. The user data in the NM PDU may be set independently of the TxRingDataAllowed flag using the NmOsek\_SetUserData() function.



Let

```
uint8 osekStatus;
```

and the call of

```
NmOsek_GetStatus(channel, &osekStatus)
```

returns `E_OK` for a valid channel index `channel`. Then a status bit in `osekStatus` is set if and only if

```
((osekStatus & bitMask) == bitMask)
```

evaluates as true where `bitMask` is being a mask for a certain flag from Table 2-15.

In the following example the Active flag shall be evaluated.



#### Example

```
...
uint8 netState;
NetworkHandleType channel = 0u;

NmOsek_GetStatus(channel, &netState);

if ((netState & 0x04 /* Active */) == 0x04)
{...;}
else
{...;}
...
```



#### Note

The state of NmOsek is updated each time a NM message is received or transmitted, i.e. usually in ISR-context. Please be aware of this behavior if the application uses multiple decisions to get a result (e.g. multiple nested "if").

### 2.10.4 Gateway Functionality: Remote Sleep Indication and Cancellation

In order to synchronize networks it might be necessary to get an indication whether no more network nodes require bus communication. This is the so-called 'Remote Sleep Indication'. The start of the remote sleep indication is performed by NmOsek by calling:

```
void Nm_RemoteSleepIndication(const NetworkHandleType)
```

In case a NM message is received with `Sleep.ind = Sleep.ack = 0` or a Skipped Alive message is transmitted the function

```
void Nm_RemoteSleepCancellation(const NetworkHandleType)
```

is called. This is also the case when the NmAwake state is entered or a NM message with `Sleep.ind = 1` and `Sleep.ack = 1` has been received but the application still needs bus communication.

The current Remote Sleep Status may be determined by the function

```
Std_ReturnType NmOsek_CheckRemoteSleepIndication(const  
NetworkHandleType, boolean * const nmRemoteSleepIndPtr)
```

(refer to section 4.2.1.17 for more details about this function).

## 2.10.5 User Data Handling

User data may be set in the NM TX PDU and user data may be retrieved from NM RX PDUs.

### 2.10.5.1 Setting User Data in the NM TX PDU

There are two possibilities to set user data in the NM TX PDU. Setting User Data via the Com module is simpler if the signals inside the user data bytes are modelled in the database.

Both possibilities do not conform to [5] since user data can also be set when the logical ring is still unstable.

#### 2.10.5.1.1 Setting User Data via the Com Module

This is an optional feature and can be used if 'Com User Data Support' is turned ON in the Nm module configuration.

OSEK NM supports the possibility to write the NM user data via Com signals (refer to [12] for further details). Therefore the signals have to be provided within one or two extra I-PDUs in the configuration. If two I-PDUs are used, the first I-PDU contains user data for wake-up (ALIVE) messages, e.g. reasons why an 'ALIVE' message is sent. The second I-PDU contains user data for awake (RING, LIMPHOME) messages, so for instance reasons for sending these messages are given.

The signals can for instance be set in the Application by using Rte\_Write\_<Signal name>() or by a CDD by using Com\_SendSignal().

If only one user data I-PDU is used, its contents is used in ALIVE; RING and LIMPHOME messages.

Note that there needs to be a Routing Path in PduR for each user data I-PDU.

#### 2.10.5.1.2 Setting User Data via Nm\_SetUserData()

Inside the NM Interface configuration (cf. [7]) the 'User Data Enabled' switch might be activated (but it could also not be activated if your configuration prohibits this feature).

If activated, the NM Interface function Nm\_SetUserData() can be used to set user data bytes (unless 'Com User Data Support' is ON).

All bytes that are not used as system bytes (see section 2.8.2 for details) can be customized by using Nm\_SetUserData().



#### Caution

The user data set by Nm\_SetUserData() may be lost after certain types of NM messages are being sent by the own node in the network.

### 2.10.5.2 Getting User Data from the NM RX PDU

Nm\_GetUserData() and Nm\_GetPduData() may be used to retrieve user data bytes from the most recently received NM PDU. It makes sense to call these functions in the context

of `Nm_PduRxIndication()` or `Nm_NmOsek_PduRxIndication()` (there are possibilities to configure callout functions in Nm when this function is being called).

Within the context of `Nm_NmOsek_PduRxIndication()`, it is even easier to access the PDU data bytes: just use the pointer to the `PduInfoType` data structure.

### 2.10.6 Node Identifier Services

The local source node identifier can be retrieved by the service

```
Std_ReturnType NmOsek_GetLocalNodeIdentifier (
    const NetworkHandleType nmChannelHandle,
    uint8* const nmNodeIdPtr )
```

The source node identifier from the last received message can be retrieved by the service

```
Std_ReturnType NmOsek_GetNodeIdentifier (
    const NetworkHandleType nmChannelHandle,
    uint8* nmNodeIdPtr )
```

Refer to section 4.2.1.10 and 4.2.1.11 for details about these functions.

### 2.10.7 NM PDU Receive Indication

The NM Interface is notified about the reception of an NM message by the optional function

```
Nm_NmOsek_PduRxIndication ( const NetworkHandleType
                             nmChannelHandle, const
                             PduInfoType * const pduInfo )
```

or if this function is not configured, it is also possible to be notified by the optional function

```
Nm_PduRxIndication ( const NetworkHandleType nmChannelHandle )
```

OSEK NM notifies the callback directly to the NM Interface in context of the function `NmOsek_RxIndication()`.



#### Caution

`NmOsek_RxIndication()` might be called on interrupt level so

`Nm_NmOsek_PduRxIndication()` / `Nm_PduRxIndication()` might be called on interrupt level as well.

The implementation of `Nm_NmOsek_PduRxIndication()` / `Nm_PduRxIndication()` should consume as few runtime as possible.

A burst of `Nm_NmOsek_PduRxIndication()` / `Nm_PduRxIndication()` function calls may occur on ECU startup when all nodes in the network are sending their Alive messages almost at the same time.

### 2.10.8 State Change Notifications

If the 'State Change Ind Enabled' feature is activated in the NM Interface configuration, the NmOsek uses the `Nm_StateChangeNotification()` service to notify the application about state changes.

In configurations where 'State Change Ind Enabled' is activated NmOsek calls:

```
Nm_StateChangeNotification(nmChannelHandle, oldState, newState)
```

This function is being called whenever a state change from one state to a different state from Nm\_StateType (details about this type can be found in [7]) occurs.

If the 'State Change Ind Enabled' feature is activated a function is required whose name may be configured in the NM Interface configuration [7]. If the Vector MICROSAR BswM is available, State Change Notifications can be used for handling the activation and deactivation of RX/TX messages ('Communication Control'; not to be confused with the following chapter) that functionality is implemented in this module. If not the application needs to implement this function (see 'UL\_Nm\_StateChangeNotification' in [7] for the function prototype).

### 2.10.9 Communication Control

In order to support ISO 14229 Communication Control Service \$28 the network management has a message transmission control status, which allows disabling the transmission of NM messages. For this purpose the function

```
Std_ReturnType NmOsek_DisableCommunication (  
    const NetworkHandleType nmChannelHandle )
```

can be called. The transmission of NM messages will be stopped within the next NmOsek\_MainFunction() call.

The NM PDU transmission ability is enabled again by the service

```
Std_ReturnType NmOsek_EnableCommunication (  
    const NetworkHandleType nmChannelHandle )
```



#### Caution

An ECU shall not shut down if the NM PDU transmission ability is disabled.



#### Note

The functions NmOsek\_DisableCommunication() and NmOsek\_EnableCommunication() are only available if the 'Com Control Enabled' feature is activated in the NM Interface configuration.



#### Note

The function NmOsek\_DisableCommunication() transits to the NmPassive, the NmOsek\_EnableCommunication() function transits to the NmActive state. Refer to section 2.3.7 for further details.

### 2.10.10 NM Coordinator Support

NmOsek channels can be coordinated by the NM Coordinator. Therefore it supports the Remote Sleep Indication interfaces as described in section 2.10.4.

**Caution**

NmOsek channels can currently only be actively coordinated by NM Coordinator. The following descriptions always assume that NmOsek is being coordinated actively.

In NmNormal, the coordinated shutdown takes  $T_{RingTyp} + T_{WaitBusSleep}$  after the transmission of the NM RING message with Sleep Indication bit set to 1, since the next node will send its NM RING message with Sleep Acknowledge bit set to 1 after  $T_{RingTyp}$ .

In NmLimpHome, the coordinated shutdown takes  $T_{RingMax} + T_{WaitBusSleep}$  after the transmission of the NM LIMPHOME message with Sleep Indication bit set to 1.

However, the setting 'Generic Bus Nm Shutdown Time' in Nm requires a static value for the coordinated shutdown duration, regardless of any state in NmOsek (NmNormal or NmLimpHome).

Therefore, NmOsek requires the setting in 'Generic Bus Nm Shutdown Time' to be  $T_{RingMax} + T_{WaitBusSleep} + 2 * t_{NmCycle}$ , where  $t_{NmCycle}$  denotes the 'Main Function Period' of the Nm module.  $2 * t_{NmCycle}$  is being used as temporal safety interval so that NmOsek\_NetworkRelease() will be called shortly before the message transmission. Since the duration of the shutdown procedure in NmOsek depends on whether it is in NmNormal or NmLimpHome, this time has been chosen as the "least common multiple" of both times.

Additionally, NmOsek supports the notification of a Synchronization Point via Nm\_SynchronizationPoint(). A Synchronization Point notifies the NM Coordinator that global shutdown for a coordination cluster can be started after the notification if all coordinated channels in the cluster are ready to sleep. Since the latter information is only available in the Nm module, not in the NmOsek module, Nm\_SynchronizationPoint() is called cyclically by NmOsek.

More precisely, NmOsek chooses the moment to call Nm\_SynchronizationPoint() in a way that NM Interface will call NmOsek\_NetworkRelease() shortly before the transmission of its NM RING/LIMPHOME message with the Sleep Indication bit set to 1. This happens

- > when the logical ring is stable once during each ring cycle
- > when NmOsek is in NmLimpHome once between each transmission of the LIMPHOME message.

The following subsections describe background knowledge about the calculation of the moments when the notification shall occur.

#### 2.10.10.1 Synchronization Point Notification in NmNormal

The notification NmNormal occurs when the logical ring is stable once per ring cycle. One ring cycle shall be defined as the interval between the transmission of two RING messages by the ECU. This interval is also called 'Ring Cycle Time'.

NmOsek can determine from the configuration how long it takes after a notification of Nm\_SynchronizationPoint() that the NM Coordinator calls NmOsek\_NetworkRelease(). This interval is called 'Sync Point Network Release Interval Time'.

The Nm\_SynchronizationPoint() function needs to be called once during one whole ring cycle. After it has been called, the function call of NmOsek\_NetworkRelease() by Nm shall

follow after 'Sync Point Network Release Interval Time'. After the function call of NmOsek\_NetworkRelease(), it always shall take 'Generic Bus Nm Shutdown Time'<sup>4</sup> until Bus Sleep Mode is entered.

Therefore, Nm\_SynchronizationPoint() is called  $(T_{RingMax} + t_{NmCycle} - T_{RingTyp} + \text{'Sync Point Network Release Interval Time'}) \% (\text{'Previous Ring Cycle Time'})$  before NmOsek sends its own NM message. For the calculation of 'Previous Ring Cycle Time', see below.

Note that the next message after the own NM message with Sleep Indication Bit 1 will be the one that triggers the transition to Wait Bus Sleep (RING message with Sleep Acknowledge bit set to 1).

So it needs to be calculated for the moment when Nm\_SynchronizationPoint() is called.

The 'Ring Cycle Time' is measured for each Ring Cycle by NmOsek.

On the next TX RING, the currently measured 'Ring Cycle Time' is used as the 'Previous Ring Cycle Time' mentioned above and the Ring Cycle Time measurement is restarted.

That means that the time between each TX Ring is measured and saved as 'Previous Ring Cycle Time'.

This needs to be calculated at run-time, because the number of nodes in the logical ring is dynamic and the interval between each RING message sent by each ECU may vary.

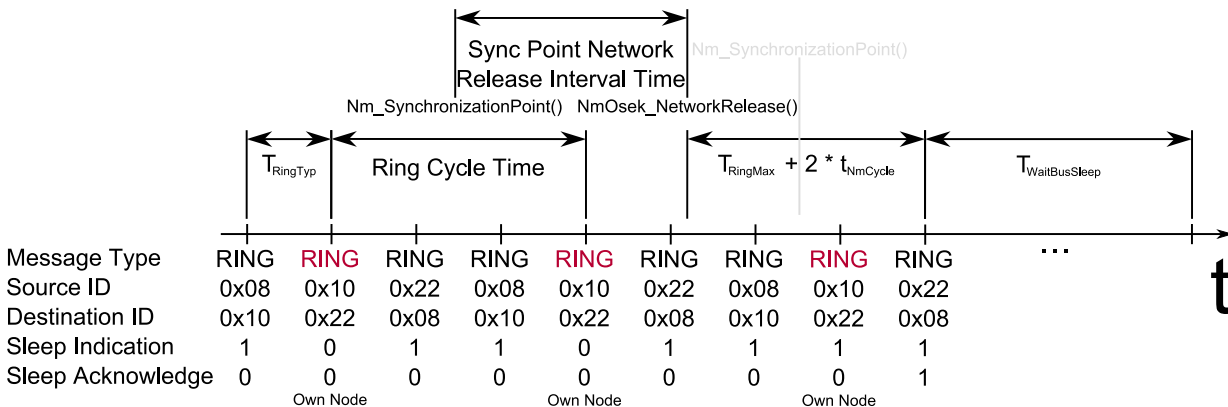


Figure 2-6 Synchronization Point Notification in NmNormal

In Figure 2-6, an example for a logical ring with three participants, nodes 0x08, 0x10 (own node) and 0x22 is provided. In this example, Ring Cycle Time is approximately  $3 * T_{RingTyp}$ . Let 'Sync Point Network Release Interval Time' be  $28 * t_{NmCycle}$ ,  $T_{RingTyp} := 10 * t_{NmCycle}$ ,  $T_{RingMax} := 26 * t_{NmCycle}$ . Since the logical ring shall be stable for some time, the 'Previous Ring Cycle Time' shall have the same value as the 'Ring Cycle Time', i.e.  $3 * T_{RingTyp}$ . The moment where Nm\_SynchronizationPoint() shall be calculated calculates as  $((26 + 1 - 10 + 28) * t_{NmCycle}) \% (30 * t_{NmCycle}) = 15 * t_{NmCycle}$ , i.e. it will be called 15 Nm 'Main Function Periods' before the transmission of the RING message in each ring cycle.

#### 2.10.10.2 Synchronization Point Notification in NmLimpHome

The notification of Nm\_SynchronizationPoint() in NmLimpHome occurs at a moment, so that the 'Sync Point Network Release Interval Time' elapses shortly before the LIMPHOME message transmission. Since there is no Ring Cycle Time in NmLimpHome but the cyclical transmission of the LIMPHOME message instead, the interval of the LIMPHOME message ( $T_{Error}$ ) is used for modulo calculation. So Nm\_SynchronizationPoint() is called once between

<sup>4</sup> If 'Wait Bus Sleep Extensions' (OEM-specific feature, only available if mentioned in [6]) are turned ON and NmOsek is in NmNormal, it shall take 'Generic Bus Nm Shutdown Time' -  $(T_{ErrorWaitBusSleep} - T_{WaitBusSleep})$

two LIMPHOME messages, 'Sync Point Network Release Interval Time' %  $T_{Error}$  before the transmission of the LIMPHOME message.

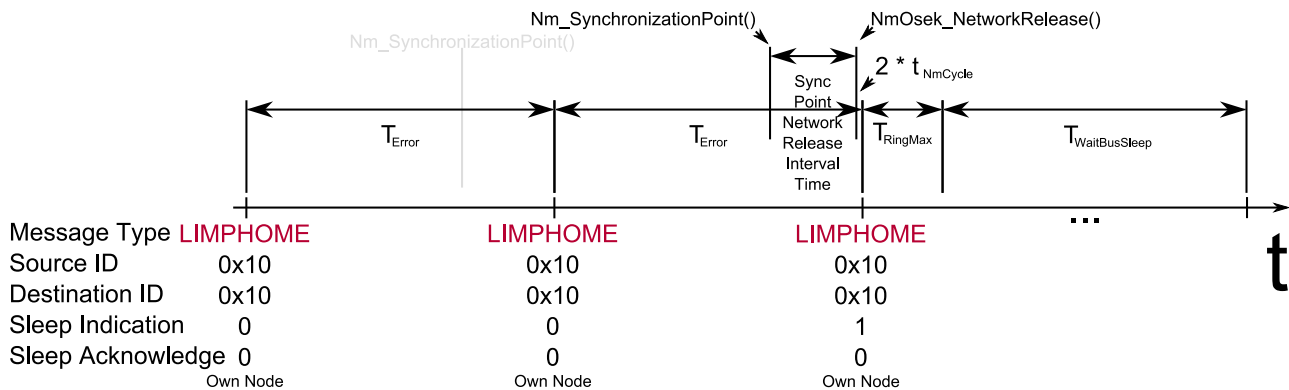


Figure 2-7 Synchronization Point Notification in NmLimpHome

Figure 2-7 depicts an example for NmLimpHome. Let 'Sync Point Network Release Interval Time' be  $28 * t_{NmCycle}$ ,  $T_{Error}$  be  $100 * t_{NmCycle}$ ,  $T_{RingMax}$  be  $26 * t_{NmCycle}$ . So  $Nm\_SynchronizationPoint()$  is called  $(28 * t_{NmCycle}) \% T_{Error} = 28 * t_{NmCycle}$  before the LIMPHOME message transmission.

#### 2.10.11 First Message shall be NM Message

NmOsek provides the possibility to ensure that the NM message shall be the first message on the bus when waking up from Bus Sleep.

This can be achieved by also using the Nm State Change Notifications to enable/disable TX I-PDU Groups (i.e. TX Application Messages in Com). If the feature is enabled, State Change Notifications to Repeat Message, Normal Operation or Ready Sleep via  $Nm\_StateChangeNotification()$  are delayed until a NM message has been successfully transmitted or received. If one of these states shall be entered from Bus Sleep or Prepare Bus Sleep, the current state is changed to Wait Startup instead if neither a NM message has been received nor transmitted.

The TX I-PDU-Group handling (typically in BswM) can then be adapted to turn TX I-PDU-Groups on only if the Nm State is not equal to Bus Sleep, Prepare Bus Sleep, Wait Startup or BusOff (if 'Bus Off Notification Enabled' is turned ON).



#### Note

In previous NmOsek versions, if the feature 'First Message shall be NM Message' is turned ON in the configuration, the usage of the CanIf setting 'Pn Wakeup Tx Pdu Filter Support' was mandatory. This is no longer required for channels with NmOsek as only NM algorithm.

This feature requires the 'Tx Confirmation Timeout Time' to be defined (see also chapter 2.8.3).



If message transmission of the NM message does not work for some reason, only NM Interface is being notified by `Nm_TxTimeoutException()` (also refer to chapter 2.8.3).

If NmOsek shall no longer wait until a successful reception or transmission of the NM message, the service function `NmOsek_CancelWaitForTxConfOrRxInd()` can be called. Then, state changes to Repeat Message, Normal Operation or Ready Sleep are also permitted.

#### 2.10.12 Handle of the TX Deadline Monitoring by Com

NmOsek provides the possibility to handle the TX Deadline Monitoring for Nm-Pdus via Com. This feature requires 'Com User Data Support' to be turned ON in Nm and the 'Tx Pdu User Data' to be defined.

If the feature is active, the 'Tx Confirmation Timeout Time' of NmOsek is not considered for message observation (see chapter 2.8.3). Therefore, the parameter `NmOsekTxConfirmationTimeoutTime` is not considered for Tx-Deadline Monitoring and there will be no function call of `Nm_TxTimeoutException()`.

Instead, a signal or a signal group inside the I-PDU which is referenced by `NmOsekTxUserDataPduRef` can be used for the timeout notification.

Similar to the sequence provided in chapter 2.8.3, the procedure is as follows:

- > OSEK NM issues a function call of `Com_TriggerIPDUSend()`.
- > OSEK NM issues a transmission request for a NM message (`CanIf_Transmit()`).
- > The Com module will handle the TX Deadline Monitoring.
- > If a message transmission is successful on the bus, Com is informed about the successful transmission via `PduR_NmOsekTxConfirmation()`.
- > Shortly before another NM message transmit request is issued, `Com_TriggerIPDUSend()` is called again.

This feature can only be used if 'Passive Mode Enabled' is turned OFF in Nm and if 'First Message Shall Be Nm Message' (see chapter 2.10.11) is turned OFF in NmOsek.



## 3 Integration

This chapter gives necessary information for the integration of the MICROSAR NmOsek into an application environment of an ECU.

### 3.1 Scope of Delivery

The delivery of the NmOsek contains the files which are described in the chapters 3.1.1 and 3.1.2:

#### 3.1.1 Static Files

| File Name    | Source Code Delivery | Description                                                                                                                                                                                                |
|--------------|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NmOsek.c     | ■                    | This is the source file of NmOsek. It contains all implementations of the services provided by the module as well as the whole Network Management algorithm.<br>The user <b>must not</b> change this file. |
| NmOsek.h     | ■                    | This is the header file for NmOsek APIs.<br>The user <b>must not</b> change this file.                                                                                                                     |
| NmOsek_Cbk.h | ■                    | This is the header file for NmOsek APIs.<br>The user <b>must not</b> change this file.                                                                                                                     |

Table 3-1 Static files

#### 3.1.2 Dynamic Files

The dynamic files are generated by the configuration tool GENy.

| File Name      | Description                                                                                              |
|----------------|----------------------------------------------------------------------------------------------------------|
| NmOsek_Cfg.c   | This is the pre-compile variant configuration source file.<br>The user <b>must not change</b> this file. |
| NmOsek_PBCfg.c | This is the post-build variant configuration source file.<br>The user <b>must not change</b> this file.  |
| NmOsek_Cfg.h   | This is the configuration header file.<br>The user <b>must not change</b> this file.                     |

Table 3-2 Generated files

### 3.2 Include Structure

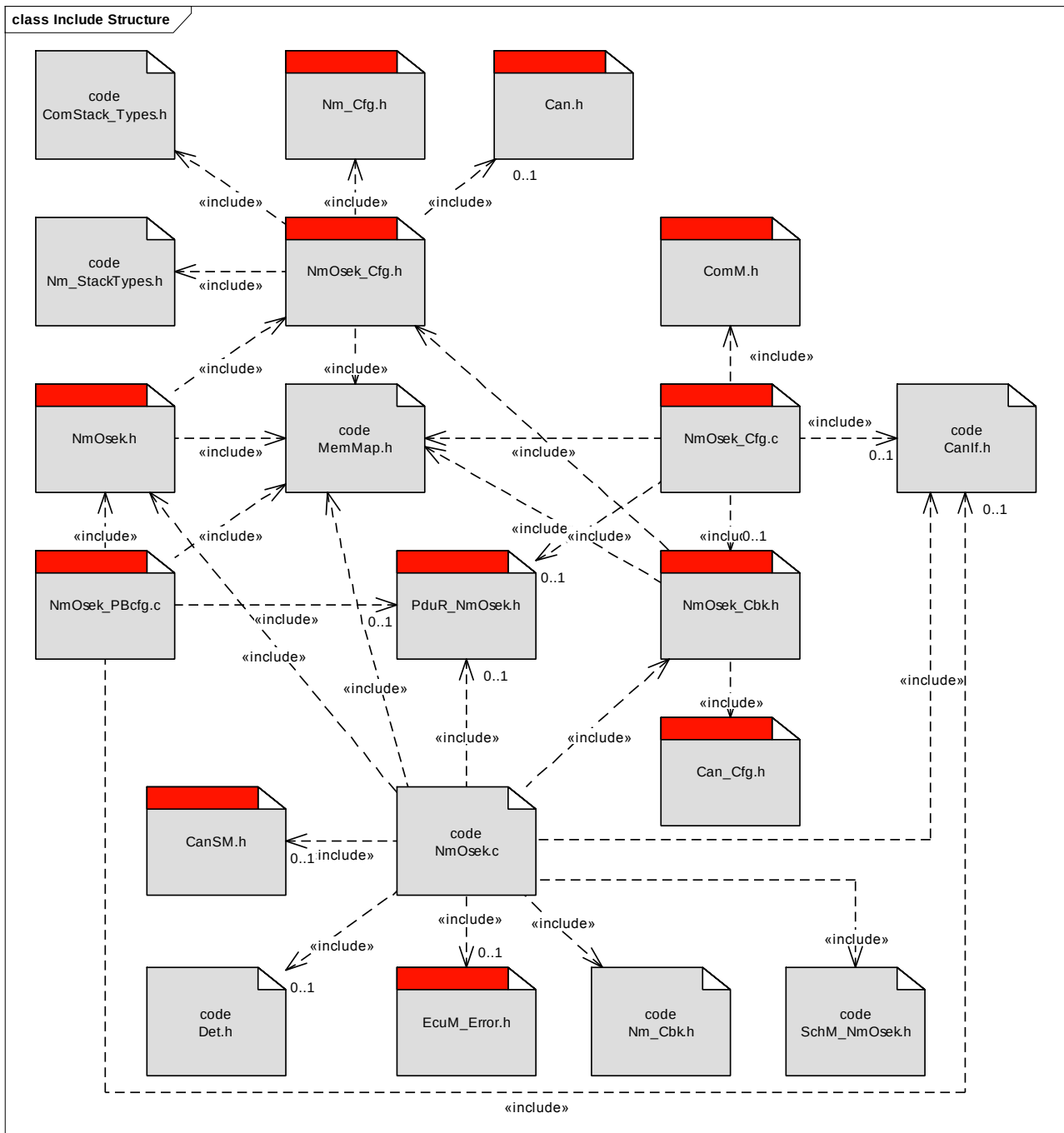


Figure 3-1 Include structure

### 3.3 Critical Sections

The AUTOSAR standard provides with the BSW Scheduler (SchM) a BSW module, which handles entering and leaving critical sections.

Critical sections are supported by using the BSW Scheduler inside the RTE.

NmOsek calls the following functions when entering a critical section  $i = 0, \dots, 5$ :

```
void SchM_Enter_NmOsek_NMOSEK_EXCLUSIVE_AREA_i()
```

When the critical section is left the following functions are called by NmOsek:

```
void SchM_Exit_NmOsek_NMOSEK_EXCLUSIVE_AREA_i()
```

The critical section (exclusive area) codes passed to these functions have to be defined and mapped to corresponding interrupt locks by the BSW Scheduler. Details which section needs what kind of interrupt lock are provided in the following section. For more information about the BSW Scheduler please refer to [9].

### 3.3.1 Critical Section Codes

In case of the BSW Scheduler is used for critical sections NmOsek uses several critical section codes which must lead to corresponding interrupt locks. The mapping is described in the following table:

| Critical Section Define | Interrupt Lock                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NMOSEK_EXCLUSIVE_AREA_0 | No interruption by any interrupt is allowed. Therefore this section must always lock global interrupts.                                                                                                                                                                                                                                                                                                                                                                                                     |
| NMOSEK_EXCLUSIVE_AREA_1 | <p>No interruption of NmOsek_NetworkRelease(), NmOsek_DisableCommunication() and NmOsek_EnableCommunication() by NmOsek_MainFunction() allowed.</p> <p>This means that global interrupts have to be used for this section only if NmOsek_NetworkRelease(), NmOsek_DisableCommunication() and NmOsek_EnableCommunication() can be interrupted by the following function:</p> <ul style="list-style-type: none"> <li>&gt; NmOsek_MainFunction()</li> </ul> <p>Otherwise no interrupt locks are necessary.</p> |
| NMOSEK_EXCLUSIVE_AREA_2 | <p>No interruption of NmOsek_PassiveStartUp() by NmOsek_MainFunction() allowed.</p> <p>This means that global interrupts must be locked if NmOsek_PassiveStartUp() can be interrupted by the following function:</p> <ul style="list-style-type: none"> <li>&gt; NmOsek_MainFunction()</li> </ul> <p>Otherwise no interrupt locks are necessary.</p>                                                                                                                                                        |
| NMOSEK_EXCLUSIVE_AREA_3 | <p>No interruption of NmOsek_SetUserData() by NmOsek_MainFunction() allowed</p> <p>This means that global interrupts must be locked if NmOsek_SetUserData() can be interrupted by the following functions:</p> <ul style="list-style-type: none"> <li>&gt; NmOsek_MainFunction()</li> </ul> <p>Otherwise no interrupt locks are necessary.</p>                                                                                                                                                              |
| NMOSEK_EXCLUSIVE_AREA_4 | <p>No interruption of NmOsek_GetUserData() by NmOsek_RxIndication() or NmOsek_GetPduData() allowed</p> <p>This means that global interrupts must be locked if NmOsek_GetUserData() can be interrupted by the following functions:</p> <ul style="list-style-type: none"> <li>&gt; NmOsek_RxIndication()</li> <li>&gt; NmOsek_GetPduData()</li> </ul> <p>Otherwise no interrupt locks are necessary.</p>                                                                                                     |
| NMOSEK_EXCLUSIVE_AREA_5 | <p>No interruption of NmOsek_GetPduData() allowed</p> <p>This means that global interrupts must be locked if NmOsek_GetPduData() can be interrupted by the following functions:</p> <ul style="list-style-type: none"> <li>&gt; NmOsek_RxIndication()</li> <li>&gt; NmOsek_GetUserData()</li> </ul> <p>Otherwise no interrupt locks are necessary.</p>                                                                                                                                                      |

Table 3-3 Critical Section Codes

## 4 API Description

For an interfaces overview please see Figure 1-2.

### 4.1 Type Definitions

The types defined by the NmOsek are described in this chapter.

| Type Name         | C-Type | Description                       | Value Range                                                                                                                                                                                                                                                |
|-------------------|--------|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NmOsek_ActionType | uint8  | Type of the internal action state | NMOSEK_NO_ACTION (0x00)<br>NmOsek_MainFunction() call has no effect for the channel                                                                                                                                                                        |
|                   |        |                                   | NMOSEK_ACTION_START (0x01)<br>Transient state that decides whether NmNormal or NmLimpHome should be entered                                                                                                                                                |
|                   |        |                                   | NMOSEK_ACTION_WAIT_SENDTOKEN (0x02)<br>Token at own node, wait before sending a Ring message                                                                                                                                                               |
|                   |        |                                   | NMOSEK_ACTION_WAIT_RINGMSG (0x03)<br>Token at another node, waiting for Ring message                                                                                                                                                                       |
|                   |        |                                   | NMOSEK_ACTION_LH_WAIT_LIMPHOME_TX (0x04)<br>NmRun in NmLimpHome state                                                                                                                                                                                      |
|                   |        |                                   | NMOSEK_ACTION_LH_PREPARE_SLEEP (0x05)<br>NmPrepSleep in NmLimpHome state                                                                                                                                                                                   |
|                   |        |                                   | NMOSEK_ACTION_GO_BUSSLEEP (0x06)<br>NmWaitBusSleep state                                                                                                                                                                                                   |
|                   |        |                                   | NMOSEK_ACTION_BUS_SLEEP (0x07)<br>NmBusSleep state                                                                                                                                                                                                         |
|                   |        |                                   | NMOSEK_ACTION_START_WITH_REPEAT (0x81)<br>Same value as NMOSEK_ACTION_START plus repetition flag (0x80). When this value has been set to a state variable the actions for NMOSEK_ACTION_START are processed within the same call of NmOsek_MainFunction(). |
|                   |        |                                   |                                                                                                                                                                                                                                                            |

| Type Name | C-Type | Description | Value Range                                                                                                                                                                                                                                                                                                                            |
|-----------|--------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|           |        |             | NMOSEK_ACTION_LH_PREPARE_SLEEP_WITH_REPEAT (0x85)<br><b>Same value as</b><br>NMOSEK_ACTION_LH_PREPARE_SLEEP plus repetition flag (0x80).<br><b>When this value has been set to a state variable the actions for</b><br>NMOSEK_ACTION_LH_PREPARE_SLEEP <b>are processed within the</b><br><b>same call of</b><br>NmOsek_MainFunction(). |

Table 4-1 Type definitions

## NmOsek\_NodeConfigType

This is used by a variable that represents the current network configuration. i.e. a byte array that indicates which nodes in the network are present or absent (for all channels). For details refer to 2.10.2. A pointer to a variable of this type is also used in the function NmOsek\_GetConfig().

Note that this type definition is only available when the 'Use Nm Node List' feature is enabled in the NmOsek configuration.

| C-Type                                                                                          | Description       | Value Range                        |
|-------------------------------------------------------------------------------------------------|-------------------|------------------------------------|
| uint8[N]<br>with N = (Number of maximum nodes in the network / 8) * (Number of NmOsek channels) | Global state byte | 0-255 (for each byte in the array) |

Table 4-2 NmOsek\_NodeConfigType

## 4.2 Services Provided by NmOsek

### 4.2.1 Services Provided due to AUTOSAR Requirements

The following functions are implemented by NmOsek to fulfill the AUTOSAR requirement to implement a 'Generic BusNm' as CDD.

### 4.2.1.1 NmOsek\_Init

| Prototype                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| void <b>NmOsek_Init</b> (const NmOsek_ConfigType *nmOsekConfigPtr )                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                     |
| Parameter                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                     |
| nmOsekConfigPtr                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Configuration structure for initializing the module |
| Return code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                     |
| -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | -                                                   |
| Functional Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                     |
| Initialize the complete NmOsek module, i.e. all channels which are activated at configuration time are initialized.                                                                                                                                                                                                                                                                                                                                                                          |                                                     |
| Particularities and Limitations                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                     |
| <ul style="list-style-type: none"> <li>&gt; Service ID: see table 'Service IDs'</li> <li>&gt; This function has to be called before any other function (except NmOsek_InitMemory() or NmOsek_GetVersionInfo())</li> <li>&gt; NmOsek_Cfg.h contains symbolic defines (NmOsek_Config_Ptr / NmOsek_Config_&lt;Predefined Variant Name&gt;_Ptr) that can be used as configuration parameter</li> <li>&gt; This function is non-reentrant.</li> <li>&gt; This function is synchronous.</li> </ul> |                                                     |
| Expected Caller Context                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                     |
| <ul style="list-style-type: none"> <li>&gt; At system start-up on task level</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                      |                                                     |

Table 4-3 NmOsek\_Init

### 4.2.1.2 NmOsek\_PassiveStartUp

| Prototype                                                                                                                                                                                                                                                                                                     |                                                                                |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| Std_ReturnType NmOsek_PassiveStartUp(<br>const NetworkHandleType nmChannelHandle )                                                                                                                                                                                                                            |                                                                                |
| Parameter                                                                                                                                                                                                                                                                                                     |                                                                                |
| nmChannelHandle                                                                                                                                                                                                                                                                                               | Identification of the NM-channel                                               |
| Return code                                                                                                                                                                                                                                                                                                   |                                                                                |
| Std_ReturnType                                                                                                                                                                                                                                                                                                | E_OK - No error<br>E_NOT_OK - Passive startup of network management has failed |
| Functional Description                                                                                                                                                                                                                                                                                        |                                                                                |
| Passive startup of the OSEK NM. It triggers the transition from Bus-Sleep Mode (NmBusSleep) to the Network Mode (NmAwake) in Repeat Message State. This service has no effect if the current state is neither equal to Prepare Bus Sleep Mode nor equal to Bus Sleep Mode. In that case E_NOT_OK is returned. |                                                                                |
| Particularities and Limitations                                                                                                                                                                                                                                                                               |                                                                                |
| <ul style="list-style-type: none"> <li>&gt; Service ID: see table 'Service IDs'</li> <li>&gt; Called by NM Interface.</li> <li>&gt; This function is non-reentrant.</li> <li>&gt; This function is asynchronous.</li> <li>&gt; Pre-condition: NmOsek is initialized</li> </ul>                                |                                                                                |
| Expected Caller Context                                                                                                                                                                                                                                                                                       |                                                                                |
| <ul style="list-style-type: none"> <li>&gt; Can be called from interrupt or task level</li> </ul>                                                                                                                                                                                                             |                                                                                |

Table 4-4 NmOsek\_PassiveStartUp



### 4.2.1.3 NmOsek\_NetworkRequest

| Prototype                                                                                                                                                                                                                                                                                                                                                                                                         |                                                             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| Std_ReturnType NmOsek_NetworkRequest(<br>const NetworkHandleType nmChannelHandle )                                                                                                                                                                                                                                                                                                                                |                                                             |
| Parameter                                                                                                                                                                                                                                                                                                                                                                                                         |                                                             |
| nmChannelHandle                                                                                                                                                                                                                                                                                                                                                                                                   | Identification of the NM-channel                            |
| Return code                                                                                                                                                                                                                                                                                                                                                                                                       |                                                             |
| Std_ReturnType                                                                                                                                                                                                                                                                                                                                                                                                    | E_OK - No error<br>E_NOT_OK - Requesting network has failed |
| Functional Description                                                                                                                                                                                                                                                                                                                                                                                            |                                                             |
| Request the network since ECU needs to communicate on the bus.                                                                                                                                                                                                                                                                                                                                                    |                                                             |
| Particularities and Limitations                                                                                                                                                                                                                                                                                                                                                                                   |                                                             |
| <ul style="list-style-type: none"> <li>&gt; Service ID: see table 'Service IDs'</li> <li>&gt; Called by NM Interface.</li> <li>&gt; This function is non-reentrant.</li> <li>&gt; This function is asynchronous.</li> <li>&gt; Pre-condition: NmOsek is initialized</li> <li>&gt; This function is not available if the 'Passive Mode Enabled' feature is activated in the NM Interface configuration.</li> </ul> |                                                             |
| Expected Caller Context                                                                                                                                                                                                                                                                                                                                                                                           |                                                             |
| <ul style="list-style-type: none"> <li>&gt; Can be called from interrupt or task level</li> </ul>                                                                                                                                                                                                                                                                                                                 |                                                             |

Table 4-5 NmOsek\_NetworkRequest

#### 4.2.1.4 NmOsek\_NetworkRelease

| Prototype                                                                                                                                                                                                                                                                                                                                                                                                         |                                                            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| Std_ReturnType NmOsek_NetworkRelease (<br>const NetworkHandleType nmChannelHandle )                                                                                                                                                                                                                                                                                                                               |                                                            |
| Parameter                                                                                                                                                                                                                                                                                                                                                                                                         |                                                            |
| nmChannelHandle                                                                                                                                                                                                                                                                                                                                                                                                   | Identification of the NM-channel                           |
| Return code                                                                                                                                                                                                                                                                                                                                                                                                       |                                                            |
| Std_ReturnType                                                                                                                                                                                                                                                                                                                                                                                                    | E_OK - No error<br>E_NOT_OK - Releasing network has failed |
| Functional Description                                                                                                                                                                                                                                                                                                                                                                                            |                                                            |
| Release the network since ECU doesn't have to communicate on the bus.                                                                                                                                                                                                                                                                                                                                             |                                                            |
| Particularities and Limitations                                                                                                                                                                                                                                                                                                                                                                                   |                                                            |
| <ul style="list-style-type: none"> <li>&gt; Service ID: see table 'Service IDs'</li> <li>&gt; Called by NM Interface.</li> <li>&gt; This function is non-reentrant.</li> <li>&gt; This function is asynchronous.</li> <li>&gt; Pre-condition: NmOsek is initialized</li> <li>&gt; This function is not available if the 'Passive Mode Enabled' feature is activated in the NM Interface configuration.</li> </ul> |                                                            |
| Expected Caller Context                                                                                                                                                                                                                                                                                                                                                                                           |                                                            |
| <ul style="list-style-type: none"> <li>&gt; Can be called from interrupt or task level</li> </ul>                                                                                                                                                                                                                                                                                                                 |                                                            |

Table 4-6 NmOsek\_NetworkRelease

### 4.2.1.5 NmOsek\_DisableCommunication

| Prototype                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Std_ReturnType NmOsek_DisableCommunication (<br>const NetworkHandleType nmChannelHandle )                                                                                                                                                                                                                                                                                                                         |                                                                                     |
| Parameter                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                     |
| nmChannelHandle                                                                                                                                                                                                                                                                                                                                                                                                   | Identification of the NM-channel                                                    |
| Return code                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                     |
| Std_ReturnType                                                                                                                                                                                                                                                                                                                                                                                                    | E_OK - No error<br>E_NOT_OK - Disabling the NM PDU transmission ability has failed. |
| Functional Description                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                     |
| Disable the NM PDU transmission ability due to an ISO14229 Communication Control (28hex) service.                                                                                                                                                                                                                                                                                                                 |                                                                                     |
| Particularities and Limitations                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                     |
| <ul style="list-style-type: none"> <li>&gt; Service ID: see table 'Service IDs'</li> <li>&gt; Called by NM Interface.</li> <li>&gt; This function is non-reentrant.</li> <li>&gt; This function is asynchronous.</li> <li>&gt; Pre-condition: NmOsek is initialized</li> <li>&gt; This function is only available if the 'Com Control Enabled' feature is activated in the NM Interface configuration.</li> </ul> |                                                                                     |
| Expected Caller Context                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                     |
| <ul style="list-style-type: none"> <li>&gt; Can be called from interrupt or task level</li> </ul>                                                                                                                                                                                                                                                                                                                 |                                                                                     |

Table 4-7 NmOsek\_DisableCommunication

#### 4.2.1.6 NmOsek\_EnableCommunication

| Prototype                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Std_ReturnType NmOsek_EnableCommunication(<br>const NetworkHandleType nmChannelHandle )                                                                                                                                                                                                                                                                                                                           |                                                                                    |
| Parameter                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                    |
| nmChannelHandle                                                                                                                                                                                                                                                                                                                                                                                                   | Identification of the NM-channel                                                   |
| Return code                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                    |
| Std_ReturnType                                                                                                                                                                                                                                                                                                                                                                                                    | E_OK - No error<br>E_NOT_OK - Enabling the NM PDU transmission ability has failed. |
| Functional Description                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                    |
| Enable the NM PDU transmission ability due to an ISO14229 Communication Control (28hex) service.                                                                                                                                                                                                                                                                                                                  |                                                                                    |
| Particularities and Limitations                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                    |
| <ul style="list-style-type: none"> <li>&gt; Service ID: see table 'Service IDs'</li> <li>&gt; Called by NM Interface.</li> <li>&gt; This function is non-reentrant.</li> <li>&gt; This function is asynchronous.</li> <li>&gt; Pre-condition: NmOsek is initialized</li> <li>&gt; This function is only available if the 'Com Control Enabled' feature is activated in the NM Interface configuration.</li> </ul> |                                                                                    |
| Expected Caller Context                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                    |
| <ul style="list-style-type: none"> <li>&gt; Can be called from interrupt or task level</li> </ul>                                                                                                                                                                                                                                                                                                                 |                                                                                    |

Table 4-8 NmOsek\_EnableCommunication

### 4.2.1.7 NmOsek\_SetUserData

| Prototype                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                      |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <pre>Std_ReturnType NmOsek_SetUserData (     const NetworkHandleType nmChannelHandle,     const uint8 * const nmUserDataPtr )</pre>                                                                                                                                                                                                                                                                                                                                                                                |                                                                                      |
| Parameter                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                      |
| nmChannelHandle                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Identification of the NM-channel                                                     |
| nmUserDataPtr                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Pointer where the user data for the next transmitted NM message shall be copied from |
| Return code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                      |
| Std_ReturnType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | E_OK - No error<br>E_NOT_OK - Setting user data has failed                           |
| Functional Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                      |
| Set user data for NM messages transmitted next on the bus.                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                      |
| Particularities and Limitations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                      |
| <ul style="list-style-type: none"> <li>&gt; Service ID: see table 'Service IDs'</li> <li>&gt; Called by NM Interface.</li> <li>&gt; This function is non-reentrant.</li> <li>&gt; This function is synchronous.</li> <li>&gt; Pre-condition: NmOsek is initialized</li> <li>&gt; This function is only available if the 'User Data Enabled' feature is activated and neither the 'Passive Mode Enabled' feature nor the 'Com User Data Support' feature is activated in the NM Interface configuration.</li> </ul> |                                                                                      |
| Expected Caller Context                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                      |
| <ul style="list-style-type: none"> <li>&gt; Can be called from interrupt or task level</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                      |

Table 4-9 NmOsek\_SetUserData

### 4.2.1.8 NmOsek\_GetUserData

| Prototype                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| <pre>Std_ReturnType NmOsek_GetUserData (     const NetworkHandleType nmChannelHandle,     uint8 * const nmUserDataPtr )</pre>                                                                                                                                                                                                                                                                                  |                                                                                         |
| Parameter                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                         |
| nmChannelHandle                                                                                                                                                                                                                                                                                                                                                                                                | Identification of the NM-channel                                                        |
| nmUserDataPtr                                                                                                                                                                                                                                                                                                                                                                                                  | Pointer where user data out of the most recently received NM message shall be copied to |
| Return code                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                         |
| Std_ReturnType                                                                                                                                                                                                                                                                                                                                                                                                 | E_OK - No error<br>E_NOT_OK - Getting user data has failed                              |
| Functional Description                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                         |
| Get user data out of the most recently received NM message.                                                                                                                                                                                                                                                                                                                                                    |                                                                                         |
| Particularities and Limitations                                                                                                                                                                                                                                                                                                                                                                                |                                                                                         |
| <ul style="list-style-type: none"> <li>&gt; Service ID: see table 'Service IDs'</li> <li>&gt; Called by NM Interface.</li> <li>&gt; This function is non-reentrant.</li> <li>&gt; This function is synchronous.</li> <li>&gt; Pre-condition: NmOsek is initialized</li> <li>&gt; This function is only available if the 'User Data Enabled' feature is activated in the NM Interface configuration.</li> </ul> |                                                                                         |
| Expected Caller Context                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                         |
| <ul style="list-style-type: none"> <li>&gt; Can be called from interrupt or task level</li> </ul>                                                                                                                                                                                                                                                                                                              |                                                                                         |

Table 4-10 NmOsek\_GetUserData

### 4.2.1.9 NmOsek\_Transmit

| Prototype                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                              |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>Std_ReturnType NmOsek_Transmit(     PduIdType NmOsekTxPduId,     const PduInfoType *PduInfoPtr )</pre>                                                                                                                                                                                                                                                 |                                                                                                                                                                                              |
| Parameter                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                              |
| NmOsekTxPduId                                                                                                                                                                                                                                                                                                                                               | L-PDU handle of CAN L-PDU to be transmitted. This handle specifies the corresponding CAN L-PDU ID and implicitly the CAN Driver instance as well as the corresponding CAN controller device. |
| PduInfoPtr                                                                                                                                                                                                                                                                                                                                                  | Pointer to a structure with CAN L-PDU related data: DLC and pointer to CAN L-SDU buffer                                                                                                      |
| Return code                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                              |
| Std_ReturnType                                                                                                                                                                                                                                                                                                                                              | E_OK - Transmit request has been accepted Function could be called from interrupt level or from task level                                                                                   |
| Functional Description                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                              |
| NmOsek_Transmit is implemented as an empty function and shall always return E_OK.                                                                                                                                                                                                                                                                           |                                                                                                                                                                                              |
| Particularities and Limitations                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                              |
| <ul style="list-style-type: none"> <li>&gt; Service ID: see table 'Service IDs'</li> <li>&gt; Called by NM Interface.</li> <li>&gt; This function is reentrant.</li> <li>&gt; This function is synchronous.</li> <li>&gt; This function is only available if the 'Com User Data Enabled' feature is activated in the NM Interface configuration.</li> </ul> |                                                                                                                                                                                              |
| Expected Caller Context                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                              |
| <ul style="list-style-type: none"> <li>&gt; Can be called from interrupt or task level</li> </ul>                                                                                                                                                                                                                                                           |                                                                                                                                                                                              |

Table 4-11 NmOsek\_Transmit

#### 4.2.1.10 NmOsek\_GetNodeIdentifier

| Prototype                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| <pre>Std_ReturnType NmOsek_GetNodeIdentifier(<br/>    const NetworkHandleType nmChannelHandle,<br/>    uint8 * const nmNodeIdPtr )</pre>                                                                                                                                                                                                                                                          |                                                                                                               |
| Parameter                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                               |
| nmChannelHandle                                                                                                                                                                                                                                                                                                                                                                                   | Identification of the NM-channel                                                                              |
| nmNodeIdPtr                                                                                                                                                                                                                                                                                                                                                                                       | Pointer where node identifier out of the most recently received NM PDU shall be copied to                     |
| Return code                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                               |
| Std_ReturnType                                                                                                                                                                                                                                                                                                                                                                                    | E_OK - No error<br>E_NOT_OK - Getting the node identifier out of the most recently received NM PDU has failed |
| Functional Description                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                               |
| Get node identifier of the most recently received NM PDU.                                                                                                                                                                                                                                                                                                                                         |                                                                                                               |
| Particularities and Limitations                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                               |
| <ul style="list-style-type: none"><li>&gt; Service ID: see table 'Service IDs'</li><li>&gt; Called by NM Interface.</li><li>&gt; This function is reentrant.</li><li>&gt; This function is synchronous.</li><li>&gt; Pre-condition: NmOsek is initialized</li><li>&gt; This function is only available if the 'Node Id Enabled' feature is activated in the NM Interface configuration.</li></ul> |                                                                                                               |
| Expected Caller Context                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                               |
| <ul style="list-style-type: none"><li>&gt; Can be called from interrupt or task level</li></ul>                                                                                                                                                                                                                                                                                                   |                                                                                                               |

Table 4-12 NmOsek\_GetNodeIdentifier



#### 4.2.1.11 NmOsek\_GetLocalNodeIdentifier

| Prototype                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| <pre>Std_ReturnType NmOsek_GetLocalNodeIdentifier(     const NetworkHandleType nmChannelHandle,     uint8 * const nmNodeIdPtr )</pre>                                                                                                                                                                                                                                                                    |                                                                                        |
| Parameter                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                        |
| nmChannelHandle                                                                                                                                                                                                                                                                                                                                                                                          | Identification of the NM-channel                                                       |
| nmNodeIdPtr                                                                                                                                                                                                                                                                                                                                                                                              | Pointer where node identifier of the local node shall be copied to                     |
| Return code                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                        |
| Std_ReturnType                                                                                                                                                                                                                                                                                                                                                                                           | E_OK - No error<br>E_NOT_OK - Getting the node identifier of the local node has failed |
| Functional Description                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                        |
| Get node identifier configured for the local node.                                                                                                                                                                                                                                                                                                                                                       |                                                                                        |
| Particularities and Limitations                                                                                                                                                                                                                                                                                                                                                                          |                                                                                        |
| <ul style="list-style-type: none"> <li>&gt; Service ID: see table 'Service IDs'</li> <li>&gt; Called by NM Interface.</li> <li>&gt; This function is reentrant.</li> <li>&gt; This function is synchronous.</li> <li>&gt; Pre-condition: NmOsek is initialized</li> <li>&gt; This function is only available if the 'Node Id Enabled' feature is activated in the NM Interface configuration.</li> </ul> |                                                                                        |
| Expected Caller Context                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                        |
| <ul style="list-style-type: none"> <li>&gt; Can be called from interrupt or task level</li> </ul>                                                                                                                                                                                                                                                                                                        |                                                                                        |

Table 4-13 NmOsek\_GetLocalNodeIdentifier

#### 4.2.1.12 NmOsek\_RepeatMessageRequest

| Prototype                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Std_ReturnType NmOsek_RepeatMessageRequest (<br>const NetworkHandleType nmChannelHandle )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                       |
| Parameter                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                       |
| nmChannelHandle                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Identification of the NM-channel                                      |
| Return code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                       |
| Std_ReturnType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | E_NOT_OK - Transition to NMNormal Request (Repeat Message) has failed |
| Functional Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                       |
| Request state change to Repeat Message (NMNormal) State.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                       |
| Particularities and Limitations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                       |
| <ul style="list-style-type: none"> <li>&gt; Service ID: see table 'Service IDs'</li> <li>&gt; Called by NM Interface.</li> <li>&gt; This function is reentrant.</li> <li>&gt; This function is asynchronous.</li> <li>&gt; Pre-condition: NmOsek is initialized</li> <li>&gt; This function is only available if the 'Node Detection Enabled' feature is activated in the NM Interface configuration.</li> <li>&gt; This function never returns E_OK since the equivalent state for the AUTOSAR 'Repeat Message State' that has been chosen for this implementation is the period of time where the logical ring is unstable. A transition to this state cannot be forced by a function call. For details about states please refer to section 2.3.</li> <li>&gt; If there is a need to find out which nodes are available on the network at a certain time one can use the NmOsek_GetConfig() and NmOsek_CmpConfig() functions instead.</li> </ul> |                                                                       |
| Expected Caller Context                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                       |
| <ul style="list-style-type: none"> <li>&gt; Can be called from interrupt or task level</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                       |

Table 4-14 NmOsek\_RepeatMessageRequest

### 4.2.1.13 NmOsek\_GetPduData

| Prototype                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| <pre>Std_ReturnType NmOsek_GetPduData (     const NetworkHandleType nmChannelHandle,     uint8 * const nmPduDataPtr )</pre>                                                                                                                                                                                                                                                                                                                                                 |                                                                  |
| Parameter                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                  |
| nmChannelHandle                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Identification of the NM-channel                                 |
| nmPduDataPtr                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Pointer where NM PDU Data shall be copied to                     |
| Return code                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                  |
| Std_ReturnType                                                                                                                                                                                                                                                                                                                                                                                                                                                              | E_OK - No error<br>E_NOT_OK - Getting the NM PDU data has failed |
| Functional Description                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                  |
| Get the whole PDU data out of the most recently received NM message.                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                  |
| Particularities and Limitations                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                  |
| <ul style="list-style-type: none"> <li>&gt; Service ID: see table 'Service IDs'</li> <li>&gt; Called by NM Interface.</li> <li>&gt; This function is non-reentrant.</li> <li>&gt; This function is synchronous.</li> <li>&gt; Pre-condition: NmOsek is initialized</li> <li>&gt; This function is only available if the 'Node Id Enabled' feature or the 'User Data Enabled' feature is activated in the NM Interface configuration (also both of the features).</li> </ul> |                                                                  |
| Expected Caller Context                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                  |
| <ul style="list-style-type: none"> <li>&gt; Can be called from interrupt or task level</li> </ul>                                                                                                                                                                                                                                                                                                                                                                           |                                                                  |

Table 4-15 NmOsek\_GetPduData

#### 4.2.1.14 NmOsek\_GetState

| Prototype                                                                                                                                                                                                                                                                 |                                                                      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| <pre>Std_ReturnType NmOsek_GetState (     const NetworkHandleType nmChannelHandle,     Nm_StateType * const nmStatePtr,     Nm_ModeType * const nmModePtr )</pre>                                                                                                         |                                                                      |
| Parameter                                                                                                                                                                                                                                                                 |                                                                      |
| nmChannelHandle                                                                                                                                                                                                                                                           | Identification of the NM-channel                                     |
| nmStatePtr                                                                                                                                                                                                                                                                | Pointer where the state of the network management shall be copied to |
| nmModePtr                                                                                                                                                                                                                                                                 | Pointer where the mode of the network management shall be copied to  |
| Return code                                                                                                                                                                                                                                                               |                                                                      |
| Std_ReturnType                                                                                                                                                                                                                                                            | E_OK - No error<br>E_NOT_OK - Getting the NM state has failed        |
| Functional Description                                                                                                                                                                                                                                                    |                                                                      |
| Return the state and the mode of the network management.                                                                                                                                                                                                                  |                                                                      |
| Particularities and Limitations                                                                                                                                                                                                                                           |                                                                      |
| <ul style="list-style-type: none"> <li>&gt; Service ID: see table 'Service IDs'</li> <li>&gt; Called by NM Interface.</li> <li>&gt; This function is reentrant.</li> <li>&gt; This function is synchronous.</li> <li>&gt; Pre-condition: NmOsek is initialized</li> </ul> |                                                                      |
| Expected Caller Context                                                                                                                                                                                                                                                   |                                                                      |
| <ul style="list-style-type: none"> <li>&gt; Can be called from interrupt or task level</li> </ul>                                                                                                                                                                         |                                                                      |

Table 4-16 NmOsek\_GetState

#### 4.2.1.15 NmOsek\_GetVersionInfo

| Prototype                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>void NmOsek_GetVersionInfo(     Std_VersionInfoType *versioninfo )</pre>                                                                                                                                                                                                                                                                   |                                                                                                                                                       |
| Parameter                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                       |
| versioninfo                                                                                                                                                                                                                                                                                                                                     | Pointer to a variable where the version information of this module shall be stored. Function could be called from interrupt level or from task level. |
| Return code                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                       |
| -                                                                                                                                                                                                                                                                                                                                               | -                                                                                                                                                     |
| Functional Description                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                       |
| This service returns the version information of this module.                                                                                                                                                                                                                                                                                    |                                                                                                                                                       |
| Particularities and Limitations                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                       |
| <ul style="list-style-type: none"> <li>&gt; Service ID: see table 'Service IDs'</li> <li>&gt; Called by application.</li> <li>&gt; This function is reentrant.</li> <li>&gt; This function is synchronous.</li> <li>&gt; This function is only available if the 'Version Info Api' feature is activated in the NmOsek configuration.</li> </ul> |                                                                                                                                                       |
| Expected Caller Context                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                       |
| <ul style="list-style-type: none"> <li>&gt; Can be called from interrupt or task level</li> </ul>                                                                                                                                                                                                                                               |                                                                                                                                                       |

Table 4-17 NmOsek\_GetVersionInfo

#### 4.2.1.16 NmOsek\_RequestBusSynchronization

| Prototype                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|
| Std_ReturnType NmOsek_RequestBusSynchronization (<br>const NetworkHandleType nmChannelHandle )                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                      |
| Parameter                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                      |
| nmChannelHandle                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Identification of the NM-channel                     |
| Return code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                      |
| Std_ReturnType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | E_NOT_OK – Requesting Bus Synchronization has failed |
| Functional Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                      |
| Request bus synchronization.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                      |
| Particularities and Limitations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                      |
| <ul style="list-style-type: none"> <li>&gt; Service ID: see table 'Service IDs'</li> <li>&gt; Called by NM Interface.</li> <li>&gt; This function is reentrant.</li> <li>&gt; This function is synchronous.</li> <li>&gt; This function is only available if the 'Bus Synchronization Enabled' feature is activated in the NM Interface configuration.</li> <li>&gt; The implementation of this function is empty since the Bus Synchronization feature is not applicable for this implementation and thus E_NOT_OK is always returned</li> </ul> |                                                      |
| Expected Caller Context                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                      |
| <ul style="list-style-type: none"> <li>&gt; Can be called from interrupt or task level</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                      |

Table 4-18 NmOsek\_RequestBusSynchronization

#### 4.2.1.17 NmOsek\_CheckRemoteSleepIndication

| Prototype                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| <pre>Std_ReturnType NmOsek_CheckRemoteSleepIndication (     const NetworkHandleType nmChannelHandle,     boolean * const nmRemoteSleepIndPtr )</pre>                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                           |
| Parameter                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                           |
| nmChannelHandle                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Identification of the NM-channel                                          |
| nmRemoteSleepIndPtr                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Pointer where check result of remote sleep indication shall be copied to  |
| Return code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                           |
| Std_ReturnType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | E_OK - No error<br>E_NOT_OK - Checking remote sleep indication has failed |
| Functional Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                           |
| Check if remote sleep state has been entered or not.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                           |
| Particularities and Limitations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                           |
| <ul style="list-style-type: none"> <li>&gt; Service ID: see table 'Service IDs'</li> <li>&gt; Called by NM Interface.</li> <li>&gt; This function is reentrant.</li> <li>&gt; This function is synchronous.</li> <li>&gt; Pre-condition: NmOsek is initialized</li> <li>&gt; This function is only available if the 'Remote Sleep Ind Enabled' feature is activated in the NM Interface configuration.</li> <li>&gt; The variable behind nmRemoteSleep is either set to TRUE if the remote sleep indication state has been entered or to FALSE otherwise.</li> </ul> |                                                                           |
| Expected Caller Context                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                           |
| <ul style="list-style-type: none"> <li>&gt; Can be called from interrupt or task level</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                           |

Table 4-19 NmOsek\_CheckRemoteSleepIndication

#### 4.2.1.18 NmOsek\_MainFunction

| Prototype                                                                                                                                                                                                                 |   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <code>void NmOsek_MainFunction( void )</code>                                                                                                                                                                             |   |
| Parameter                                                                                                                                                                                                                 |   |
| -                                                                                                                                                                                                                         | - |
| Return code                                                                                                                                                                                                               |   |
| -                                                                                                                                                                                                                         | - |
| Functional Description                                                                                                                                                                                                    |   |
| Main function of the NmOsek which processes the OSEK NM algorithm. This function is responsible to handle all NmOsek instances.                                                                                           |   |
| Particularities and Limitations                                                                                                                                                                                           |   |
| <ul style="list-style-type: none"> <li>&gt; Service ID: see table 'Service IDs'</li> <li>&gt; Called by SCHM.</li> <li>&gt; This function is non-reentrant.</li> <li>&gt; Pre-condition: NmOsek is initialized</li> </ul> |   |
| Expected Caller Context                                                                                                                                                                                                   |   |
| <ul style="list-style-type: none"> <li>&gt; Can be called from task level</li> </ul>                                                                                                                                      |   |

Table 4-20 NmOsek\_MainFunction

### 4.2.1 Additional Services

The following functions may be used by the application.

#### 4.2.1.1 NmOsek\_InitMemory

| Prototype                                                                                                                                                                                                                      |   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <code>void NmOsek_InitMemory( void )</code>                                                                                                                                                                                    |   |
| Parameter                                                                                                                                                                                                                      |   |
| -                                                                                                                                                                                                                              | - |
| Return code                                                                                                                                                                                                                    |   |
| -                                                                                                                                                                                                                              | - |
| Functional Description                                                                                                                                                                                                         |   |
| Initialize memory so that expected start values are set. Refer to section 2.2 for further details.                                                                                                                             |   |
| Particularities and Limitations                                                                                                                                                                                                |   |
| <ul style="list-style-type: none"> <li>&gt; Service ID: (none)</li> <li>&gt; Called in start-up code.</li> <li>&gt; Has to be called with disabled global interrupts.</li> <li>&gt; This function is non-reentrant.</li> </ul> |   |
| Expected Caller Context                                                                                                                                                                                                        |   |
| <ul style="list-style-type: none"> <li>&gt; Can be called on task level</li> </ul>                                                                                                                                             |   |

Table 4-21 NmOsek\_InitMemory



### 4.2.1.2 NmOsek\_GetStatus

| Prototype                                                                                                                                                                                                                                                          |                                                                            |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| <pre>Std_ReturnType NmOsek_GetStatus (<br/>    const NetworkHandleType nmChannelHandle,<br/>    uint8 * const nmStatusPtr )</pre>                                                                                                                                  |                                                                            |
| Parameter                                                                                                                                                                                                                                                          |                                                                            |
| nmChannelHandle                                                                                                                                                                                                                                                    | Identification of the NM-channel                                           |
| nmStatusPtr                                                                                                                                                                                                                                                        | Pointer where the status byte of the network management shall be copied to |
| Return code                                                                                                                                                                                                                                                        |                                                                            |
| Std_ReturnType                                                                                                                                                                                                                                                     | E_OK - No error<br>E_NOT_OK - Getting the NM state has failed              |
| Functional Description                                                                                                                                                                                                                                             |                                                                            |
| Return the global status byte of the network management.                                                                                                                                                                                                           |                                                                            |
| Particularities and Limitations                                                                                                                                                                                                                                    |                                                                            |
| <ul style="list-style-type: none"><li>&gt; Service ID: see table 'Service IDs'</li><li>&gt; Called by application.</li><li>&gt; This function is synchronous.</li><li>&gt; This function is reentrant.</li><li>&gt; Pre-condition: NmOsek is initialized</li></ul> |                                                                            |
| Expected Caller Context                                                                                                                                                                                                                                            |                                                                            |
| <ul style="list-style-type: none"><li>&gt; Can be called on task or interrupt level</li></ul>                                                                                                                                                                      |                                                                            |

Table 4-22 NmOsek\_GetStatus

### 4.2.1.3 NmOsek\_GetConfig

| Prototype                                                                                                                                                                                                                                                                                                                                                                                              |                                                         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| <pre>void NmOsek_GetConfig(     const NetworkHandleType nmChannelHandle,     NmOsek_NodeConfigType *nmDataPtr )</pre>                                                                                                                                                                                                                                                                                  |                                                         |
| Parameter                                                                                                                                                                                                                                                                                                                                                                                              |                                                         |
| nmChannelHandle                                                                                                                                                                                                                                                                                                                                                                                        | Identification of the NM-channel                        |
| nmDataPtr                                                                                                                                                                                                                                                                                                                                                                                              | Pointer where the node configuration array is copied to |
| Return code                                                                                                                                                                                                                                                                                                                                                                                            |                                                         |
| -                                                                                                                                                                                                                                                                                                                                                                                                      | -                                                       |
| Functional Description                                                                                                                                                                                                                                                                                                                                                                                 |                                                         |
| This API copies the current node configuration into an application specific buffer.                                                                                                                                                                                                                                                                                                                    |                                                         |
| Particularities and Limitations                                                                                                                                                                                                                                                                                                                                                                        |                                                         |
| <ul style="list-style-type: none"> <li>&gt; Service ID: see table 'Service IDs'</li> <li>&gt; Called by application.</li> <li>&gt; This function is synchronous.</li> <li>&gt; This function is non-reentrant.</li> <li>&gt; Pre-condition: NmOsek is initialized</li> <li>&gt; This function is only available if the 'Use Nm Node List' feature is activated in the NmOsek configuration.</li> </ul> |                                                         |
| Expected Caller Context                                                                                                                                                                                                                                                                                                                                                                                |                                                         |
| <ul style="list-style-type: none"> <li>&gt; Can be called on task level</li> </ul>                                                                                                                                                                                                                                                                                                                     |                                                         |

Table 4-23 NmOsek\_GetConfig

#### 4.2.1.4 NmOsek\_CmpConfig

| Prototype                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| <pre>boolean NmOsek_CmpConfig(     const NetworkHandleType nmChannelHandle,     uint8 idx)</pre>                                                                                                                                                                                                                                                                                                       |                                                                                              |
| Parameter                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                              |
| nmChannelHandle                                                                                                                                                                                                                                                                                                                                                                                        | Identification of the NM-channel                                                             |
| idx                                                                                                                                                                                                                                                                                                                                                                                                    | Index of ConfigMaskTable and TargetConfigTable                                               |
| Return code                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                              |
| boolean                                                                                                                                                                                                                                                                                                                                                                                                | TRUE Configuration matches the reference<br>FALSE Configuration does not match the reference |
| Functional Description                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                              |
| This API compares the current NM node configuration with a reference configuration stored in memory.                                                                                                                                                                                                                                                                                                   |                                                                                              |
| Particularities and Limitations                                                                                                                                                                                                                                                                                                                                                                        |                                                                                              |
| <ul style="list-style-type: none"> <li>&gt; Service ID: see table 'Service IDs'</li> <li>&gt; Called by application.</li> <li>&gt; This function is synchronous.</li> <li>&gt; This function is non-reentrant.</li> <li>&gt; Pre-condition: NmOsek is initialized</li> <li>&gt; This function is only available if the 'Use Nm Node List' feature is activated in the NmOsek configuration.</li> </ul> |                                                                                              |
| Expected Caller Context                                                                                                                                                                                                                                                                                                                                                                                |                                                                                              |
| <ul style="list-style-type: none"> <li>&gt; Can be called on task level</li> </ul>                                                                                                                                                                                                                                                                                                                     |                                                                                              |

Table 4-24 NmOsek\_CmpConfig

#### 4.2.1.5 NmOsek\_CancelWaitForTxConfOrRxInd

| Prototype                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Std_ReturnType NmOsek_CancelWaitForTxConfOrRxInd (<br>const NetworkHandleType nmChannelHandle)                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                          |
| Parameter                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                          |
| nmChannelHandle                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Identification of the NM-channel                                                                                                                                                         |
| Return code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                          |
| boolean                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | E_OK – No error<br>E_NOT_OK - The flag to indicate that NmOsek shall no longer wait for a TxConfirmation or RxIndication cannot be set in the current state or incorrect nmChannelHandle |
| Functional Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                          |
| <p>Cancel the process of NmOsek to wait for a message confirmation or message reception<br/>Call this function to let NmOsek no longer wait for one of these events to happen:</p> <ul style="list-style-type: none"><li>- NM PDU reception</li><li>- NM PDU confirmation</li></ul> <p>If this function has been called, neither a NM PDU reception nor a NM PDU confirmation needs to happen to prevent state changes to NM_STATE_REPEAT_MESSAGE, NM_STATE_NORMAL_OPERATION or NM_STATE_READY_SLEEP.</p> |                                                                                                                                                                                          |
| Particularities and Limitations                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                          |
| <ul style="list-style-type: none"><li>&gt; Service ID: see table 'Service IDs'</li><li>&gt; Called by application.</li><li>&gt; This function is synchronous.</li><li>&gt; This function is reentrant.</li><li>&gt; Pre-condition: NmOsek is initialized</li><li>&gt; This function is only available if the 'First Message Shall Be Nm Message' feature is activated in the NmOsek configuration.</li></ul>                                                                                              |                                                                                                                                                                                          |
| Expected Caller Context                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                          |
| <ul style="list-style-type: none"><li>&gt; Can be called on task or interrupt level</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                          |

Table 4-25 NmOsek\_CancelWaitForTxConfOrRxInd

### 4.3 Services Used by NmOsek

In the following table services provided by other components, which are used by the NmOsek are listed. For details about prototype and functionality refer to the documentation of the providing component.

| Component | API                              |
|-----------|----------------------------------|
| CANIF     | CanIf_Transmit <sup>5</sup>      |
| COM       | Com_TriggerIPDUSend <sup>6</sup> |
| DET       | Det_ReportError <sup>7</sup>     |

<sup>5</sup> Only used if 'Passive Mode Enabled' is turned OFF in the NM Interface configuration.

<sup>6</sup> Only used if 'Tx Deadline Monitoring in Com Enabled' is turned ON

<sup>7</sup> Only used if 'Dev Error Detect' is turned ON.

| Component  | API                                                           |
|------------|---------------------------------------------------------------|
| ECUM       | EcuM_BswErrorHook <sup>8</sup>                                |
| NM         | Nm_NetworkStartIndication                                     |
|            | Nm_NetworkMode                                                |
|            | Nm_PrepareBusSleepMode                                        |
|            | Nm_BusSleepMode                                               |
|            | Nm_StateChangeNotification <sup>9</sup>                       |
|            | Nm_PduRxIndication <sup>10</sup>                              |
|            | Nm_NmOsek_PduRxIndication <sup>11</sup>                       |
|            | Nm_RemoteSleepIndication <sup>12</sup>                        |
|            | Nm_RemoteSleepCancellation <sup>12</sup>                      |
|            | Nm_SynchronizationPoint <sup>13</sup>                         |
|            | Nm_TxTimeoutException <sup>14</sup>                           |
| PDUR       | PduR_NmOsekTriggerTransmit <sup>15</sup>                      |
|            | PduR_NmOsekTxConfirmation <sup>15</sup>                       |
| SCHM (RTE) | SchM_Enter_NmOsek_NMOSEK_EXCLUSIVE_AREA_i<br>with i = 0,...,5 |
|            | SchM_Exit_NmOsek_NMOSEK_EXCLUSIVE_AREA_i<br>with i = 0,...,5  |

Table 4-26 Services used by the NmOsek

## 4.4 Callback Functions

This chapter describes the callback functions that are implemented by the NmOsek and can be invoked by other modules.

### 4.4.1 Callback Functions That Exist Due To AUTOSAR Requirements

The following callback functions are called by CANIF.

<sup>8</sup> Only used if NmOsek is configured in the Configuration Variant VARIANT-POST-BUILD-LOADABLE (also possible in combination with Variant Support).

<sup>9</sup> Only used if the feature 'State Change Ind Enabled' is activated in the NM Interface configuration.

<sup>10</sup> Only used if the feature 'Bus Nm Specific Pdu Rx Indication Enabled' is deactivated and the feature 'Pdu Rx Indication Enabled' is activated in the NM Interface configuration.

<sup>11</sup> Only used if the feature 'Bus Nm Specific Pdu Rx Indication Enabled' is activated in the NM Interface configuration.

<sup>12</sup> Only used if the feature 'Remote Sleep Ind Enabled' is activated in the NM Interface configuration.

<sup>13</sup> Only used if the feature 'Coordinator Support Enabled' is activated in the NM Interface configuration and the 'Synchronizing Network' setting is turned ON for at least one NmChannelConfig container that belongs to a NmOsek channel

<sup>14</sup> Only used if the parameter 'Tx Confirmation Timeout Time' is defined and 'Passive Mode Enabled' is turned OFF in the NM Interface configuration.

<sup>15</sup> Only used if 'User Data Tx Pdu's are configured and 'Com User Data Support' is turned ON in the NM Interface configuration

#### 4.4.1.1 NmOsek\_TxConfirmation

| Prototype                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| <pre>void NmOsek_TxConfirmation(     PduIdType nmOsekTxPduId )</pre>                                                                                                                                                                                                                                                                                                                                                                                                     |                                               |
| Parameter                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                               |
| nmOsekTxPduId                                                                                                                                                                                                                                                                                                                                                                                                                                                            | CANIF Tx PDU ID of the transmitted NmOsek PDU |
| Return code                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                               |
| -                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | -                                             |
| Functional Description                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                               |
| This service confirms a previous successfully processed CAN transmit request.                                                                                                                                                                                                                                                                                                                                                                                            |                                               |
| Particularities and Limitations                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                               |
| <ul style="list-style-type: none"> <li>&gt; Service ID: see table 'Service IDs'</li> <li>&gt; Called by CAN Interface.</li> <li>&gt; This function is synchronous.</li> <li>&gt; This function is reentrant (but not for the same channel determined by nmOsekTxPduId).</li> <li>&gt; Pre-condition: NmOsek is initialized</li> <li>&gt; This function is not available if the 'Passive Mode Enabled' feature is activated in the NM Interface configuration.</li> </ul> |                                               |
| Expected Caller Context                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                               |
| <ul style="list-style-type: none"> <li>&gt; Can be called on task or interrupt level</li> </ul>                                                                                                                                                                                                                                                                                                                                                                          |                                               |

Table 4-27 NmOsek\_TxConfirmation

#### 4.4.1.2 NmOsek\_RxIndication

| Prototype                                                                                                                                                                                                                                                                |                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| <pre>void NmOsek_RxIndication(<br/>    PduIdType nmOsekRxPduId,<br/>    const uint8 *canSduPtr,<br/>    const Can_IdType canId )</pre>                                                                                                                                   |                                              |
| Parameter                                                                                                                                                                                                                                                                |                                              |
| nmOsekRxPduId                                                                                                                                                                                                                                                            | Identification of the network through PDU-ID |
| canSduPtr                                                                                                                                                                                                                                                                | Pointer to received SDU                      |
| canId                                                                                                                                                                                                                                                                    | CAN Identifier of the received SDU           |
| Return code                                                                                                                                                                                                                                                              |                                              |
| -                                                                                                                                                                                                                                                                        | -                                            |
| Functional Description                                                                                                                                                                                                                                                   |                                              |
| This service indicates a successful reception of a received NM message to the NmOsek after passing all filters and validation checks.                                                                                                                                    |                                              |
| Particularities and Limitations                                                                                                                                                                                                                                          |                                              |
| <ul style="list-style-type: none"><li>&gt; Service ID: see table 'Service IDs'</li><li>&gt; Called by CAN Interface.</li><li>&gt; This function is synchronous.</li><li>&gt; This function is non-reentrant.</li><li>&gt; Pre-condition: NmOsek is initialized</li></ul> |                                              |
| Expected Caller Context                                                                                                                                                                                                                                                  |                                              |
| <ul style="list-style-type: none"><li>&gt; Can be called on task or interrupt level</li></ul>                                                                                                                                                                            |                                              |

Table 4-28 NmOsek\_RxIndication

#### 4.4.2 Additional Callback Functions

The following callback functions have been added to support transitions to/from LimpHome due to notifications about BusOff occurrences/recoveries. These callback functions are implemented by NmOsek.

### 4.4.2.1 NmOsek\_CanSM\_BusOffBegin

| Prototype                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| <pre>void NmOsek_CanSM_BusOffBegin (     NetworkHandleType NetworkHandle,     uint8 *OnlineDelayCyclesPtr )</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                        |
| Parameter                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                        |
| NetworkHandle                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | system channel index                                                                   |
| OnlineDelayCyclesPtr                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Pointer to the location where the number of delay cycles shall be stored.<br>(ignored) |
| Return code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                        |
| -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | -                                                                                      |
| Functional Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                        |
| BusOff occurrence notification callback function.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                        |
| Particularities and Limitations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                        |
| <ul style="list-style-type: none"> <li>&gt; Service ID: see table 'Service IDs'</li> <li>&gt; Called by CANSIM if configured as 'Bus Off Begin' function in the CANSIM configuration. Can also be called by a CDD.</li> <li>&gt; This function is asynchronous.</li> <li>&gt; This function is reentrant.</li> <li>&gt; Pre-condition: NmOsek is initialized</li> <li>&gt; This function is only available if the 'BusOff Notification Enabled' feature is activated in the NmOsek configuration.</li> <li>&gt; If it is called by CANSIM, the function prototype is provided by CanSM.</li> <li>&gt; Refer to the CANSIM Technical Reference [8] for more information about the BusOff notification feature.</li> </ul> |                                                                                        |
| Expected Caller Context                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                        |
| <ul style="list-style-type: none"> <li>&gt; Can be called on task or interrupt level</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                        |

Table 4-29 NmOsek\_CanSM\_BusOffBegin



#### 4.4.2.2 NmOsek\_CanSM\_BusOffEnd

| Prototype                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <pre>void NmOsek_CanSM_BusOffEnd(     NetworkHandleType NetworkHandle )</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                      |
| Parameter                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                      |
| NetworkHandle                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | system channel index |
| Return code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                      |
| -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | -                    |
| Functional Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                      |
| BusOff recovery notification callback function.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                      |
| Particularities and Limitations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                      |
| <ul style="list-style-type: none"> <li>&gt; Service ID: see table 'Service IDs'</li> <li>&gt; Called by CANSIM if configured as 'Bus Off End' function in the CANSIM configuration. Can also be called by a CDD.</li> <li>&gt; This function is asynchronous.</li> <li>&gt; This function is reentrant.</li> <li>&gt; Pre-condition: NmOsek is initialized</li> <li>&gt; This function is only available if the 'BusOff Notification Enabled' feature is activated in the NmOsek configuration.</li> <li>&gt; If it is called by CANSIM, the function prototype is provided by CanSM.</li> <li>&gt; Refer to the CANSIM Technical Reference [8] for more information about the BusOff notification feature.</li> </ul> |                      |
| Expected Caller Context                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                      |
| <ul style="list-style-type: none"> <li>&gt; Can be called on task or interrupt level</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                      |

Table 4-30 NmOsek\_CanSM\_BusOffEnd

## 5 AUTOSAR Standard Compliance

### 5.1 Deviations

There are no known deviations to the AUTOSAR Standard (when NmOsek is regarded as the CanNm module and APIs similar to those of the CanNm) (so called Generic BusNm concept).

### 5.2 Additions/ Extensions

#### 5.2.1 Providing Advanced Status Information

In addition to the AUTOSAR-like function NmOsek\_GetState() the function NmOsek\_GetStatus() can be used for retrieval of internal status information.

Refer to section 2.10.3 for more information about this feature.

#### 5.2.2 Controlling the Transition to the Bus Sleep State and Vice Versa

The OSEK NM algorithm is used for controlling these transitions instead of an AUTOSAR standard algorithm (like the algorithm of CanNm).

Refer to section 2.7 for details.

#### 5.2.3 Determination and Monitoring of the Network Configuration

The network configuration, i.e. a list of the nodes that are participating inside the network, can be determined by using the functions NmOsek\_GetConfig() and NmOsek\_CmpConfig().

Refer to section for 2.10.2 further details.

### 5.3 Limitations

Since NmOsek is a Complex Device Driver there are no limitations.

## 6 Glossary and Abbreviations

### 6.1 Glossary

| Term                 | Description                                                                                                |
|----------------------|------------------------------------------------------------------------------------------------------------|
| BusNm                | A generic term for lower layer components of the AUTOSAR NM module that implement bus-specific NM behavior |
| DaVinci Configurator | Generation tool for MICROSAR components                                                                    |

Table 6-1 Glossary

### 6.2 Abbreviations

| Abbreviation | Description                                                                                                                                             |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| API          | Application Programming Interface                                                                                                                       |
| AUTOSAR      | Automotive Open System Architecture                                                                                                                     |
| BSW          | Basic Software                                                                                                                                          |
| CANIF        | AUTOSAR CAN Interface                                                                                                                                   |
| CanNm        | AUTOSAR CAN Network Management                                                                                                                          |
| CANSM        | AUTOSAR CAN State Manager                                                                                                                               |
| CDD          | Complex Device Driver                                                                                                                                   |
| DEM          | Diagnostic Event Manager                                                                                                                                |
| DET          | Development Error Tracer                                                                                                                                |
| ECU          | Electronic Control Unit                                                                                                                                 |
| FrNm         | AUTOSAR FlexRay Network Management                                                                                                                      |
| Gw           | Gateway                                                                                                                                                 |
| HIS          | Hersteller Initiative Software                                                                                                                          |
| ID           | Identifier                                                                                                                                              |
| ISR          | Interrupt Service Routine                                                                                                                               |
| J1939Nm      | AUTOSAR SAE J1939 Network Management                                                                                                                    |
| LinNm        | AUTOSAR LIN Network Management                                                                                                                          |
| MICROSAR     | Microcontroller Open System Architecture (the Vector AUTOSAR solution)                                                                                  |
| Msg          | Message                                                                                                                                                 |
| NM           | Network Management                                                                                                                                      |
| Nm           | AUTOSAR Network Management Interface                                                                                                                    |
| NmOsek       | Network Management Osek (this component)                                                                                                                |
| OEM          | Original Equipment Manufacturer                                                                                                                         |
| OSEK         | Offene Systeme und deren Schnittstellen für die Elektronik im Kraftfahrzeug (Open Systems and the Corresponding Interfaces for Automotive Electronics.) |

|       |                                |
|-------|--------------------------------|
| PDU   | Protocol Data Unit             |
| RTE   | AUTOSAR Runtime Environment    |
| SCHM  | AUTOSAR Schedule Manager       |
| SWS   | Software Specification         |
| UdpNm | AUTOSAR UDP Network Management |

Table 6-2 Abbreviations

## 7 Contact

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